Information Acquisition and the Value of Public Information in Network Games

Jiacheng Fan*

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Abstract

We study information acquisition and the value of public information in a beauty contest game on a directed network. Agents choose the precision of private information before taking actions that reflect adaptation and heterogeneous coordination motives. More precise public information crowds out private acquisition more sharply at more central positions. This endogenous response reduces the social value of public information relative to holding the precision of private information fixed. The reduction is amplified when more central agents also exert substantial influence on others. More precise public information can remain socially detrimental despite efficient acquisition, but necessarily improves welfare when information use is also efficient. We characterize policy interventions that implement the socially optimal allocations.

Keywords: Network games; Information acquisition; Social value of public information; Beauty contests.

JEL classification: C72, D62, D82, D83, D85.

*Lingnan College, Sun Yat-sen University. E-mail: fanjch7@mail2.sysu.edu.cn. The author thanks Xuesong Huang, Qian Jiao, Xueheng Li, and Jianhao Lin for helpful comments. This work was supported by the National Natural Science Foundation of China (Grant No. 723B2024). All errors are the author's own.

1 Introduction

Public information is of paramount importance for decentralized decisions with networked social interactions. In order to stabilize financial markets, central banks communicate assessments of macro-financial conditions among institutions linked by counterparty exposures. In order to guide pricing and production decisions, statistics bureaus publish official projections of aggregate demand and inflation for firms connected through supply-chain linkages. In order to coordinate complex organizations, corporate managers disseminate organization-wide forecasts across interdependent business units. In each setting, the direction and strength of these linkages create asymmetric patterns of strategic dependence and influence. Participants also acquire private information through risk assessments, market research, or customer surveys. Accordingly, the welfare effects of public information depend crucially on how network position shapes both incentives for information acquisition and the consequences of individual actions for others.

The practical importance of public information has given rise to an extensive literature investigating its economic consequences, both theoretically (Morris and Shin, 2002) and empirically (Lustenberger and Rossi, 2022). A central insight of the literature is that more precise public information need not improve social welfare when individual decisions exhibit strategic complementarities. The underlying mechanism is that private coordination incentives lead decision makers to place excessive weight on public signals relative to what is socially desirable. When public information is sufficiently imprecise, the resulting over-coordination may be harmful enough to warrant withholding the information altogether. Subsequent studies have extended the welfare analysis along two dimensions. One strand endogenizes the information structure by allowing decision makers to acquire information at a cost (Chahrour, 2014; Ui, 2014) or to allocate limited attention across different signals (Chen et al., 2015). Another strand incorporates network heterogeneity, examining how the asymmetries of social interactions shape information acquisition and use (Myatt and Wallace, 2019) or change the welfare implications of public information (Murayama, 2026).

Several questions remain at the intersection of these strands. Who acquires more private information in equilibrium, and whose acquisition departs from socially optimal levels? When public information affects private information acquisition, how does network position shape the individual and aggregate welfare consequences? Can socially optimal acquisition ensure that more precise public information increases welfare, or does the answer depend on whether information use is also efficient?

In this paper, we address these questions in a network model with endogenous private information acquisition. We characterize equilibrium acquisition and use of information and

investigate the role of network position in explaining the discrepancy between private and social incentives to acquire information. We then use this characterization to examine how the social value of public information is affected by endogenous private information acquisition in network settings. Finally, we compare the welfare effects under two socially optimal benchmarks that both feature efficient acquisition but differ in whether information use is efficient or remains at equilibrium.

We conduct our analysis in a finite-player version of the beauty contest game with a general interaction network. The payoff structure follows Myatt and Wallace (2019), while we consider a more specialized information structure with endogenous private information acquisition. Specifically, agents choose their actions to balance two objectives: adapting to an unknown fundamental state and coordinating with their neighbors. The network records the direction and intensity of these coordination motives. Information about the fundamental follows a standard Gaussian signal structure, with a common public signal observed by all agents and a private signal observed only by the individual agent. The precision of public information is exogenously given, whereas agents choose the precision of their private signals through costly information acquisition. Acquisition decisions are simultaneous and unobserved by other agents. After observing their own private signals and the public signal, agents use this information to estimate the fundamental and form expectations about their neighbors' actions. Then they simultaneously take actions.

To illustrate the model primitives, consider financial analysts forecasting a firm's future earnings. Analysts devote time and resources to proprietary research (Brown et al., 2015), while the firm releases public corporate disclosures that serve as a common source of information (Barron et al., 2017). They seek to produce accurate forecasts, while career concerns can also encourage conformity with their peers' assessments (Hong et al., 2000). The interaction network captures the importance each analyst attaches to matching particular peers. These incentives need not be reciprocal: one analyst may focus considerable attention on aligning her forecast with those of prominent colleagues whose assessments depend little on her own.

We begin by characterizing equilibrium acquisition and use of information. Following Colombo and Femminis (2008) and Myatt and Wallace (2019), we adopt linear acquisition costs to obtain a closed-form characterization. We show that when acquisition costs are sufficiently low, the game admits a unique linear equilibrium in which every agent acquires private information with strictly positive precision. The acquisition of private information and the responsiveness of actions to public information depend on Katz–Bonacich centrality (Katz, 1953; Bonacich, 1987). This measure aggregates an agent's direct coordination motives toward neighbors and the indirect motives for higher-order alignments arising from those neighbors' dependence on others. Public information provides a common focal point for such interdependent actions.

Therefore, agents with higher centrality place more weight on the public information and acquire less precise private information.

Our first result documents that more precise public information *crowds out* acquisition of private information throughout the population. The strength of this substitution, however, varies systematically with network position: the reduction in the precision of private information acquired in equilibrium is proportional to the agent's Katz–Bonacich centrality. That is, central agents who acquire less precise private information in equilibrium also reduce their acquisition more sharply when the precision of public information increases. Intuitively, a more precise public signal helps agents estimate the fundamental and anticipate their neighbors' actions, reducing the marginal return to acquiring more precise private information. At more central positions, individual decisions depend more heavily on the actions of others, reinforcing the coordination advantage of public information and amplifying the crowding-out effect on private information acquisition. This heterogeneous crowding-out response has important distributional implications for the value of public information.

We then analyze the equilibrium welfare effects of public information with endogenous acquisition of private information, focusing on how network structure shapes these effects. We evaluate individual payoffs and utilitarian social welfare ex ante, net of information acquisition costs. Our next result establishes necessary and sufficient conditions under which more precise public information reduces individual payoffs and social welfare. An agent can be harmed by an increase in the precision of public information when she is relatively less central than the neighbors she follows. Because these neighbors are more responsive to the public signal, shifts in their actions sharpen the agent's trade-off in reconciling her adaptation and coordination motives. We derive a characterization for this detrimental effect in terms of the agent's Katz–Bonacich centrality relative to a weighted aggregate of her neighbors' centralities.

Endogenous information acquisition introduces an additional welfare channel through the allocation of private information across network positions. When public information becomes more precise, agents acquire less precise private information, with the reduction being larger at more central network positions. Hence, the same neighbors who already respond more aggressively to public information also substitute more sharply away from private information. Their actions consequently become more reliant on the common signal than under exogenous private information, thereby amplifying the distortion transmitted to agents who follow them. This endogenous response further reduces the private value of public information relative to holding fixed the precision of private information. Therefore, providing more accurate public information that would otherwise benefit the individual can instead leave her worse off.

At the aggregate level, we characterize precisely when more accurate public information reduces social welfare under endogenous private information acquisition. As in the exogenous

benchmark (Murayama, 2026), the condition depends jointly on agents' Katz–Bonacich centralities and weighted in-degrees.[1] The key distinction is that endogenous acquisition creates a novel distributional effect through heterogeneous crowding out of private information. More precise public information changes the allocation of private information across the network, displacing acquisition whose benefits to others are not fully internalized. The associated welfare loss is exacerbated when high centrality coincides with high in-degree at the same position: agents who reduce acquisition most sharply also exert substantial influence on others. A core–periphery structure illustrates that the crowding-out channel can reverse a positive social value of public information in the absence of endogenous acquisition. In the analyst application, this corresponds to an improvement in the quality of corporate disclosure benefiting a tightly connected core of analysts while harming peripheral analysts who seek to align their forecasts with the core.

We also study how the discrepancy between private and social incentives in information acquisition and use shapes the social value of public information in networks. To do so, we compare the equilibrium allocation with two socially optimal benchmarks. The first jointly optimizes information acquisition and use, whereas the second optimizes acquisition subject to agents' equilibrium use of information. Following Angeletos and Pavan (2007), information remains dispersed in both benchmarks: each agent's action depends only on her own private signal and the public signal. This comparison allows us to identify separately the inefficiencies arising from the acquisition and the use of information, and thus informs policy design.

Our third result shows that equilibrium acquisition is inefficiently low under either benchmark, both at the individual and aggregate levels. Relative to the socially optimal allocation, every agent with positive weighted in-degree acquires less precise private information, and total acquisition in society is strictly lower. The reason is that agents choose the precision of private information according to its private return without internalizing the benefits their information generates for those who follow them. More precise private information brings an agent's action closer to the fundamental, allowing followers to match it with smaller deviations from the fundamental in their own actions. The planner internalizes these network externalities and therefore assigns a higher social return to private information at positions that are more influential to others.

Having characterized the socially optimal benchmarks, we ask whether correcting inefficient information acquisition alone is sufficient to guarantee a positive social value of public information. When both information acquisition and use are determined efficiently, more precise

[1] Weighted in-degree measures the total weight other agents place on matching an agent's action, accounting for both the strength of their coordination motives and the importance they attach to her. It thus captures an agent's direct social influence on others' actions.

public information necessarily increases welfare on every admissible network. By contrast, public information can remain detrimental if information acquisition is socially optimal but information use remains at equilibrium. The reason is that efficient acquisition of information can mitigate the distortion in decentralized use of information by reallocating the precision of private information across network positions, but it cannot eliminate the wedge between private and social coordination incentives. This remaining distortion creates a sharp contrast with the crowding out that occurs both in equilibrium and when information acquisition and use are jointly efficient. When only acquisition is optimized, more precise public information increases total acquisition of private information if and only if it reduces social welfare. This is because the planner seeks to mitigate the detrimental effects of public information by expanding the acquisition of private information.

The comparison between the two social optima suggests two distinct policy interventions. A Pigouvian tax on realized actions makes agents internalize the coordination externalities for others, and the optimal tax schedule implements efficiency in both information acquisition and use. Alternatively, targeted subsidies for private information acquisition can implement the social optimum under equilibrium use of information. To provide further intuition, we derive a simple subsidy rule that targets agents according to their influence, as measured by weighted in-degree. More influential agents receive larger subsidies because improving the accuracy of private information at these positions generates larger benefits for others. Both interventions raise welfare relative to the decentralized equilibrium, but correcting acquisition incentives alone does not necessarily make more precise public information socially desirable.

The paper is organized as follows. Section 1.1 expands on the related literature, and Section 2 presents the model setup. Section 3 characterizes the equilibrium acquisition and use of information and studies the value of public information in general networks with endogenous information acquisition. Section 4 analyzes two socially optimal allocations and compares the corresponding welfare effects. Section 5 concludes.

1.1 Related Literature

Our paper contributes to the literature on the social value of public information in coordination games. Following Morris and Shin (2002), this literature examines when more precise public information can reduce welfare. Angeletos and Pavan (2007) relate the value of information to the discrepancy between equilibrium and efficient use of information, while Ui and Yoshizawa (2015) characterize the welfare effects of information in symmetric quadratic games. Other contributions examine the extent of public information dissemination (Cornand and Heinemann, 2008) and the interaction between disclosure and policy intervention (James and Lawler, 2011). We investigate these welfare questions with endogenous acquisition of private information

and heterogeneous coordination motives, identifying how network position shapes both the distribution of individual payoffs and the aggregate value of public information.

The closest comparison is Murayama (2026), which studies the welfare consequences of public information in heterogeneous networks while taking the precision of private information as given. Important questions remain when private information acquisition responds endogenously to public disclosure. How does more precise public information reshape the acquisition of private information across network positions, and how does this adjustment alter the private and social value of public information? We address these questions by allowing agents to choose the precision of private information at a cost. Improvements in public information then change the allocation of private information across the network. We show that the endogenous response strictly reduces the value of public information relative to the exogenous information benchmark and can overturn an otherwise positive welfare effect.

Our analysis also relates to research on endogenous information acquisition and attention allocation. Coordination motives shape the selection of information sources (Hellwig and Veldkamp, 2009; Myatt and Wallace, 2012) and the allocation of attention between public and private signals (Chen et al., 2015). Improvements in public information can consequently displace the acquisition of private information, with welfare implications that depend on acquisition costs and incentives (Colombo and Femminis, 2008; Ui, 2014). Colombo et al. (2014) distinguish the efficiency of information acquisition under efficient and equilibrium use of information and relate the welfare contribution of crowding out to acquisition inefficiency. We examine this relation across heterogeneous network positions, where agents differ in both the response of acquisition to public information and the benefits that private information generates for others.

Our paper also connects to the network literature on how strategic interdependence shapes equilibrium behavior, leadership, and conformity (Ballester et al., 2006; Zhou and Chen, 2015; Ushchev and Zenou, 2020). While these contributions assume complete information, a growing literature introduces incomplete information and examines how network structure interacts with uncertainty and beliefs about others' behavior (de Martí and Zenou, 2015; Golub and Morris, 2020; Leister et al., 2022). Our analysis belongs to this latter strand and is particularly close to studies that endogenize information acquisition. For instance, Myatt and Wallace (2019) examine the acquisition and use of information under asymmetric coordination motives, showing that more central agents spend less on information when agents acquire the same set of sources. Leister (2020) characterizes inefficiencies in the acquisition and use of information and examines policies that publicize information investments. We complement these works by combining the welfare analysis of public information with endogenous private information acquisition in networks.

A separate strand of the literature treats networks as structures through which agents communicate or obtain information from peers (Calvó-Armengol and de Martí, 2009; Calvó-Armengol et al., 2015; Herskovic and Ramos, 2020). Connections in these settings determine the transmission or observation of information. In our model, by contrast, the network represents directed coordination incentives rather than information access. Network structure therefore shapes the incentives to acquire and use information even though information is not transmitted across links. These strategic dependencies affect the private and social value of public information.

Finally, our policy analysis relates to research on improving incentives under dispersed information (Angeletos and Pavan, 2009) and targeting resources and incentives in networks with complementarities and social norms (Demange, 2017; Belhaj and Deroïan, 2019; Galeotti et al., 2020; Meng, 2025). We consider subsidies for verifiable purchases of private information, with purchases and reimbursements kept confidential. Our result indicates that optimal subsidies are targeted toward more influential agents, reflecting the larger external benefits their private information confers on followers. Appropriately chosen subsidies implement the social optimum under equilibrium use of information and raise welfare relative to the *laissez-faire* equilibrium. These subsidies correct incentives to acquire information while leaving the inefficiency in information use unresolved, and thus the social value of public information can remain negative.

2 Model

We study a two-stage Bayesian game on a directed interaction network. A finite population of agents first acquires costly private information and then takes coordinated actions under quadratic-loss payoffs. The interaction network operates through coordination losses and the resulting action responses; it does not determine signal observation or communication.

2.1 Payoffs and Network Structure

The economy contains $n \geq 2$ agents indexed by $i \in N \equiv \{1, \ldots, n\}$. Each agent chooses an action $a_i \in \mathbb{R}$ to align her behavior with an unknown fundamental $\theta \in \mathbb{R}$ and with the actions of her neighbors. Following Myatt and Wallace (2019), agent i's payoff is given by

$$u_i(\mathbf{a}, \theta; \boldsymbol{\alpha}, \mathbf{G}) = -(1-\alpha_i) \underbrace{(a_i - \theta)^2}_{\text{adaptation}} - \alpha_i \sum_{j \neq i} g_{ij} \underbrace{(a_i - a_j)^2}_{\text{coordination}}, \tag{1}$$

where $\mathbf{a} \equiv (a_i)_{i \in N} \in \mathbb{R}^n$ is the profile of individual actions, α_i measures agent i's aggregate coordination motive, and $\mathbf{G} \equiv (g_{ij})_{i,j \in N}$ represents the interaction network. In the analysis below, we focus on the case of *strategic complements*, in which agents prefer to match their

neighbors' actions. We therefore maintain the assumption that $0 \leq \alpha_i < 1$ for every $i \in N$ and $\alpha_i > 0$ for at least one $i \in N$, so that each agent retains a positive adaptation motive and strategic interaction remains nontrivial. The vector $\boldsymbol{\alpha} \equiv (\alpha_i)_{i \in N}$ collects the agents' aggregate coordination motives.

Each entry $g_{ij} \in \mathbb{R}_+$ of the adjacency matrix $\mathbf{G}$ measures how strongly agent i wants her action to track agent j's action. To simplify the analysis, we assume that the matrix has no self-loops ($g_{ii} = 0$) and is row-stochastic ($\sum_{j \neq i} g_{ij} = 1$).[2] The directed edge points from i to j when $g_{ij} > 0$, meaning that agent i seeks to match agent j. For each agent $i \in N$, define the out-neighborhood $N_i^{\text{out}} \equiv \{j \in N : g_{ij} > 0\}$ and the in-neighborhood $N_i^{\text{in}} \equiv \{j \in N : g_{ji} > 0\}$. In this paper, an agent $j \in N_i^{\text{out}}$ is referred to as a *leader* of agent i, and conversely, an agent $j \in N_i^{\text{in}}$ is referred to as a *follower* of agent i.

The payoff in (1) combines adaptation and coordination losses. In the analyst application, the former captures the deviation of an analyst's forecast from the firm's fundamental, whereas the latter captures disagreement with the forecasts of the peers she follows. Thus, α_i measures agent i's overall coordination motive, while g_{ij} allocates this motive across individual neighbors. With $\text{diag}(\boldsymbol{\alpha})$ denoting the diagonal matrix whose ith diagonal element is α_i, we define the adjacency matrix adjusted for the strengths of coordination motives as

$$\widehat{\mathbf{G}} \equiv \text{diag}(\boldsymbol{\alpha})\mathbf{G} = (\widehat{g}_{ij})_{i,j \in N}, \tag{2}$$

where $\widehat{g}_{ij} \equiv \alpha_i g_{ij}$ is the effective coordination weight that agent i assigns to agent j. We refer to $\widehat{\mathbf{G}}$ as the *equilibrium interaction network*. Thus, $\widehat{\mathbf{G}}$ incorporates asymmetry both in whom agents seek to coordinate with and in the overall strength of their coordination motives. By construction, it is nonnegative, has no self-loops, and is row-substochastic, with $\sum_{j \neq i} \widehat{g}_{ij} = \alpha_i < 1$ for every $i \in N$.

2.2 Information Structure

The interaction network affects payoff interdependence but not the information available to agents. We adopt the Gaussian information environment of Morris and Shin (2002). The fundamental θ is drawn from an improper uniform prior on the real line. All agents observe a common public signal

$$y = \theta + \varepsilon_y, \qquad \varepsilon_y \sim \mathcal{N}(0, 1/\tau_y), \tag{3}$$

[2] The row-stochastic normalization is imposed for convenience and is not essential. The analysis applies to arbitrary nonnegative interaction matrices, with differences in row sums absorbed into the corresponding interaction intensities.

with exogenous precision $\tau_y > 0$. In addition, agent i receives a private signal

$$x_i = \theta + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, 1/\tau_i), \tag{4}$$

whose precision $\tau_i \in \mathbb{R}_+$ is endogenously chosen and may differ across agents. The noise terms $(\varepsilon_i)_{i\in N}$ and ε_y are mutually independent and independent of the fundamental θ.

We collect the precisions of private information in the vector $\boldsymbol{\tau} \equiv (\tau_i)_{i\in N} \in \mathbb{R}^n_+$ and the private signal realizations in the vector $\mathbf{x} \equiv (x_i)_{i\in N}$. For tractability, we assume that agent i chooses the precision τ_i at a linear acquisition cost, as in Colombo and Femminis (2008), Chahrour (2014), and Myatt and Wallace (2019):

$$C(\tau_i) = c\tau_i, \qquad c > 0. \tag{5}$$

When $\tau_i = 0$, the private signal is uninformative. Given the pair of signal realizations (x_i, y), agent i's posterior expectation of the fundamental is

$$\mathbb{E}[\theta \mid x_i, y] = \gamma_i x_i + (1 - \gamma_i) y,$$

where $\gamma_i \equiv \frac{\tau_i}{\tau_i + \tau_y}$ is the posterior weight on private information that measures the relative informativeness of the private signal.

2.3 Timing and Equilibrium Concept

The game proceeds in two stages. At Stage I, agents simultaneously choose the precision of their private information, with each acquisition choice and any unilateral deviation unobserved by the other agents. At Stage II, the fundamental and signals are realized. Agents observe their private signals and the public signal and simultaneously choose actions. Payoffs are then realized net of the acquisition cost incurred at Stage I.

Given the precision of private information τ_i, an action strategy $\sigma_i(\cdot, \cdot; \tau_i) : \mathbb{R}^2 \to \mathbb{R}$ for agent i is a measurable mapping from the information set (x_i, y) into a real-valued action a_i. Let Σ_i denote the set of measurable action strategies $\sigma_i : \mathbb{R}^2 \to \mathbb{R}$ available to agent i. The product space $\Sigma \equiv \prod_{i\in N} \Sigma_i$ contains all profiles of action strategies, with a typical element denoted by $\boldsymbol{\sigma} \equiv (\sigma_i)_{i\in N}$. For any vector or strategy profile, a subscript $-i$ removes agent i's component; for example, $\mathbf{x}_{-i} \equiv (x_j)_{j\neq i}$ denotes the vector of private signals of all agents other than i.

Definition 1. A vector of precisions of private information $\widehat{\boldsymbol{\tau}} \equiv (\widehat{\tau}_i)_{i\in N}$ and a profile of action strategies $\widehat{\boldsymbol{\sigma}} \equiv (\widehat{\sigma}_i)_{i\in N}$ constitute a perfect Bayesian equilibrium if the following two optimality

conditions hold for each agent $i \in N$.[3]

(i) At Stage II, for any acquired precision $\tau_i \in \mathbb{R}_+$, agent i's action strategy $\widehat{\sigma}_i(x_i, y; \tau_i)$ maximizes expected payoff conditional on the private and public signals, taking the other agents' equilibrium action strategies as given:

$$\widehat{\sigma}_i(x_i, y; \tau_i) \in \arg\max_{a_i \in \mathbb{R}} \mathbb{E}[u_i(a_i, \widehat{\boldsymbol{\sigma}}_{-i}(\mathbf{x}_{-i}, y; \widehat{\boldsymbol{\tau}}_{-i}), \theta; \boldsymbol{\alpha}, \mathbf{G}) \mid x_i, y] .$$

(ii) At Stage I, given the other agents' equilibrium acquisition of private information $\widehat{\boldsymbol{\tau}}_{-i} \in \mathbb{R}_+^{n-1}$, agent i's acquisition choice $\widehat{\tau}_i$ maximizes expected payoff net of the acquisition cost:

$$\widehat{\tau}_i \in \arg\max_{\tau_i \in \mathbb{R}_+} \{\mathbb{E}[u_i(\widehat{\sigma}_i(x_i, y; \tau_i), \widehat{\boldsymbol{\sigma}}_{-i}(\mathbf{x}_{-i}, y; \widehat{\boldsymbol{\tau}}_{-i}), \theta; \boldsymbol{\alpha}, \mathbf{G})] - c\tau_i\} .$$

For each agent $i \in N$, beliefs at Stage II are derived from the Gaussian signal structure using Bayes' rule. In the Stage-I optimality condition, expectations under any candidate precision τ_i are taken with respect to the signal distribution induced by that choice. A deviation in this precision changes her information structure and, accordingly, her optimal Stage-II action rule. Because private information acquisition is unobserved, however, other agents maintain their equilibrium beliefs and continue to follow their equilibrium choices $(\widehat{\boldsymbol{\tau}}_{-i}, \widehat{\boldsymbol{\sigma}}_{-i})$.

3 Equilibrium

We first characterize equilibrium information acquisition and use, then examine how the induced allocation of private information shapes individual payoffs and social welfare. Throughout this section, we focus on *linear* equilibria. To keep the analysis tractable and avoid corner solutions, we consider sufficiently small positive acquisition costs. In the comparative statics, we hold $(\boldsymbol{\alpha}, \mathbf{G})$ fixed and vary the precision of public information locally within the region where information acquisition and use remain interior.

3.1 Equilibrium Acquisition and Use of Information

We first characterize equilibrium information use for a fixed precision vector $\boldsymbol{\tau}$, thereby abstracting from the acquisition problem. We then determine equilibrium information acquisition using the standard envelope argument in Colombo et al. (2014).

[3] Equivalently, any perfect Bayesian equilibrium outcome can be represented by a Bayesian Nash equilibrium of the strategic form, in which each agent's pure strategy jointly specifies a precision τ_i and a contingent action rule $a_i = \sigma_i(x_i, y; \tau_i)$ for every signal realization.

At Stage II, conditional on the information set (x_i, y), sequential optimality requires

$$a_i = (1 - \alpha_i)\mathbb{E}[\theta \mid x_i, y] + \sum_{j \neq i} \widehat{g}_{ij}\mathbb{E}[a_j \mid x_i, y]. \tag{6}$$

In a linear equilibrium, the weights on private and public information sum to one, so agent i's action takes the form[4]

$$a_i = \beta_i x_i + (1 - \beta_i)y, \tag{7}$$

where the coefficient β_i measures the sensitivity of agent i's action to private information.

Substituting the linear action rule into (6) and matching coefficients on the private and public signals yields

$$\beta_i = \gamma_i(\tau_i)\Big(1 - \alpha_i + \sum_{j \neq i} \widehat{g}_{ij}\beta_j\Big), \tag{8}$$

where $\gamma_i(\tau_i) \equiv \frac{\tau_i}{\tau_i + \tau_y}$ is the posterior weight on private information for a given precision τ_i. Let $\mathbf{1} \in \mathbb{R}^n$ denote the vector of ones and $\mathbf{I}$ the $n \times n$ identity matrix, and define

$$\boldsymbol{\Gamma}(\boldsymbol{\tau}) \equiv \operatorname{diag}\big(\gamma_1(\tau_1), \ldots, \gamma_n(\tau_n)\big).$$

Writing $\boldsymbol{\beta} \equiv (\beta_i)_{i \in N}$, we can express the system of equations (8) in matrix form as

$$\boldsymbol{\beta} = \boldsymbol{\Gamma}(\boldsymbol{\tau})\big[(\mathbf{1} - \boldsymbol{\alpha}) + \widehat{\mathbf{G}}\boldsymbol{\beta}\big],$$

or equivalently,

$$\big[\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}\big]\boldsymbol{\beta} = \boldsymbol{\Gamma}(\boldsymbol{\tau})(\mathbf{1} - \boldsymbol{\alpha}).$$

Since $\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}$ is invertible,[5] we define the equilibrium mapping of information use as

$$\mathcal{B}^{\mathrm{eq}}(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G}) \equiv \big[\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}\big]^{-1}\boldsymbol{\Gamma}(\boldsymbol{\tau})(\mathbf{1} - \boldsymbol{\alpha}). \tag{9}$$

Let $\mathcal{B}_i^{\mathrm{eq}}$ denote its i-th component. The unique solution to (8) is given by

$$\beta_i = \mathcal{B}_i^{\mathrm{eq}}(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G}), \qquad \forall i \in N.$$

At Stage I, agent i chooses the precision of her private information, taking the other agents'

[4] Appendix A.1 establishes this property.

[5] Invertibility follows from the fact that $\boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}$ is nonnegative with spectral radius below one:

$$\rho\Big(\boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}\Big) \leq \Big\|\boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}\Big\|_\infty = \max_{i \in N} \gamma_i(\tau_i)\alpha_i < 1.$$

action strategies as given. Conditional on this choice, the action rule is optimized at Stage II. Applying the envelope theorem to the Stage-I problem yields the following first-order condition for an interior acquisition choice of τ_i:[6]

$$c = \left(\frac{\beta_i}{\tau_i}\right)^2. \tag{10}$$

A larger β_i makes errors in the private signal more consequential for the action, increasing the marginal value of reducing those errors through more precise information acquisition.

Combining the acquisition condition in (10) with the equilibrium information-use mapping in (9) yields the allocation characterized in Proposition 1. We express this allocation using weighted *Katz–Bonacich centrality*. For a square matrix $\mathbf{M}$, write $\rho(\mathbf{M})$ for its spectral radius.

Definition 2. For an $n \times n$ nonnegative matrix $\mathbf{M}$, a weighting vector $\mathbf{w} \in \mathbb{R}^n_+$, and a decay parameter $\delta > 0$ satisfying $\delta\rho(\mathbf{M}) < 1$, the $\mathbf{w}$-weighted Katz–Bonacich centrality vector is defined by

$$\mathbf{b}(\mathbf{M}, \delta, \mathbf{w}) = (\mathbf{I} - \delta\mathbf{M})^{-1}\mathbf{w} = \sum_{k=0}^{\infty} \delta^k \mathbf{M}^k \mathbf{w}. \tag{11}$$

The i-th component $b_i(\mathbf{M}, \delta, \mathbf{w})$ sums the weights of directed walks starting at agent i, with each walk of length k discounted by δ^k and weighted by w_j at its terminal node j. When $\mathbf{w} = \mathbf{1}$, the measure is referred to as unweighted Katz–Bonacich centrality. The spectral condition $\delta\rho(\mathbf{M}) < 1$ ensures that $(\mathbf{I} - \delta\mathbf{M})^{-1}$ exists and is nonnegative and that the Neumann series in (11) converges.

Proposition 1. *For each $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$, there exists $\bar{c} > 0$ such that, if the acquisition cost is sufficiently low ($0 < c < \bar{c}$), the game admits a unique linear equilibrium in which the weight agent i places on private information is*

$$\widehat{\beta}_i = 1 - \tau_y \sqrt{c}\, \widehat{b}_i \in (0, 1), \tag{12}$$

where $\widehat{b}_i \equiv b_i(\widehat{\mathbf{G}}, 1, \mathbf{1})$ is agent i's unweighted Katz–Bonacich centrality in the equilibrium interaction network $\widehat{\mathbf{G}}$. The precision of private information acquired in equilibrium is

$$\widehat{\tau}_i = \frac{1}{\sqrt{c}} - \tau_y \widehat{b}_i > 0. \tag{13}$$

Proof: See Appendix A.2. □

[6] For a given coefficient β_i, the component of agent i's expected loss attributable to errors in the private signal is β_i^2/τ_i. Since the action rule is chosen optimally conditional on τ_i, the envelope theorem eliminates the indirect effect through changes in the optimal action rule, and the marginal benefit of more precise private information is therefore β_i^2/τ_i^2.

Proposition 1 shows how network position shapes equilibrium acquisition and use of information. Stronger coordination motives increase the value of public information as a common signal for coordinating with neighbors. This coordination incentive depends not only on direct neighbors, but also on how their actions in turn depend on others, and therefore accumulates along interaction paths captured by Katz–Bonacich centrality. Higher centrality increases the value of the common signal for coordinating along these paths, shifting information use toward public information and reducing the marginal return to private information acquisition. As a result, more central agents acquire less precise private information. This mechanism is consistent with Myatt and Wallace (2019), where more central agents spend less on information acquisition by favoring clearer and cheaper sources of information.[7]

The low-cost condition ensures that private information remains inexpensive enough for every agent to choose strictly positive precision, including agents at highly central positions whose acquisition incentives are weakest. [8] When information acquisition becomes sufficiently costly, these agents are the first to abandon private information altogether, giving rise to corner solutions that obscure the closed-form characterization.

Corollary 1. *Suppose that the acquisition cost is sufficiently low. An increase in the precision of public information reduces the precision of private information acquired by every agent in equilibrium:*

$$\frac{\partial \widehat{\tau}_i}{\partial \tau_y} = -\widehat{b}_i < 0. \tag{14}$$

Moreover, the substitutability between public and private information is stronger for agents with higher equilibrium Katz–Bonacich centrality. For any $i, j \in N$,

$$\widehat{b}_i > \widehat{b}_j \quad \Longleftrightarrow \quad \frac{\partial \widehat{\tau}_i}{\partial \tau_y} < \frac{\partial \widehat{\tau}_j}{\partial \tau_y}. \tag{15}$$

Proof. The vector $\widehat{\mathbf{b}}$ depends only on $(\boldsymbol{\alpha}, \mathbf{G})$. Differentiating (13) with respect to τ_y gives (14). The Neumann expansion implies $\widehat{b}_i > 0$ for every i, and comparing components gives (15). □

[7] The information environment in Myatt and Wallace (2019) is more general, featuring multiple information sources with both common and idiosyncratic noise. Our public and private signals correspond to the limiting cases with no idiosyncratic and no common noise, respectively. We treat public information as freely available and allow agents to choose the precision of their private information at a common marginal cost. In Myatt and Wallace (2019), more central agents spend less on information by shifting toward clearer and cheaper sources; in our setting, they acquire less precise private information as their stronger coordination motives increase reliance on the public signal.

[8] A sufficient condition is $\bar{c} = (\tau_y \max_{i \in N} \widehat{b}_i)^{-2}$. Since the precision of private information acquired in equilibrium decreases with Katz–Bonacich centrality, this bound guarantees positive acquisition even for the most central agent, and hence $\widehat{\beta}_i > 0$ and $\widehat{\tau}_i > 0$ for all $i \in N$. As in Calvó-Armengol et al. (2015), the restriction rules out binding lower bounds on costly precision choices.

Corollary 1 extends the crowding-out result of Colombo et al. (2014) to a network setting: agents who acquire less private information because of their higher Katz–Bonacich centrality also reduce acquisition more sharply when public information becomes more precise.

3.2 The Private and Social Value of Public Information

We now evaluate the welfare consequences of public information in equilibrium. Given the precisions of private information $\boldsymbol{\tau}$ and the precision of public information τ_y, agent i's expected payoff net of her acquisition cost is

$$U_i(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G}) \equiv \mathbb{E}[u_i(\sigma(\mathbf{x}, y; \boldsymbol{\tau}), \theta; \boldsymbol{\alpha}, \mathbf{G})] - c\tau_i,$$

where $\sigma(\mathbf{x}, y; \boldsymbol{\tau}) \equiv (\sigma_i(x_i, y; \tau_i))_{i \in N}$ is the profile of individual actions. Furthermore, define

$$W(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G}) = \sum_{i \in N} U_i(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})$$

as the utilitarian welfare net of acquisition costs. We refer to $dU_i(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})/d\tau_y$ as the *private value of public information* for agent i, and to $dW(\boldsymbol{\tau}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})/d\tau_y$ as the *social value of public information*.

Before characterizing the value of public information, we introduce the concept of weighted in-degree. Agent i's *weighted in-degree* in the interaction network is $d_i^{\text{in}} \equiv \sum_{j \neq i} \widehat{g}_{ji} = \sum_{j \neq i} \alpha_j g_{ji}$. It sums the effective weights $\widehat{g}_{ji}$ on incoming links. Because $\widehat{g}_{ji}$ is the coordination motive that agent j assigns to matching agent i's action, weighted in-degree measures the aggregate coordination motive of i's followers, $j \in N_i^{\text{in}}$, toward her action.

Proposition 2. *For each $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$, there exists $\bar{c} > 0$ such that, if the acquisition cost is sufficiently low ($0 < c < \bar{c}$), the following results hold with endogenous acquisition of private information:*

(i) *The private value of public information for agent $i \in N$ is negative if and only if*

$$\widehat{b}_i^2 < \alpha_i \sum_{j \neq i} g_{ij} \widehat{b}_j (\widehat{b}_j - 1). \tag{16}$$

(ii) *The social value of public information is negative if and only if*

$$\sum_{i \in N} \left[(1 - d_i^{\text{in}}) \widehat{b}_i^2 + d_i^{\text{in}} \widehat{b}_i \right] < 0. \tag{17}$$

Proof: See Appendix A.4. □

Part (i) shows that public information can be detrimental to agent i when she is relatively less central than the neighbors she follows, as captured by a comparison between her own Katz–Bonacich centrality and a weighted function of her neighbors' centralities. The result reflects a tension between adaptation and coordination. More precise public information helps agent i assess both the fundamental and the actions she seeks to match, while simultaneously altering the behavior of the neighbors she follows. More central neighbors are more responsive to the public signal and, with endogenous acquisition, substitute more strongly away from private information. Network feedback consequently makes their actions increasingly tied to the common signal, amplifying the distortion induced by its noise. This amplified distortion can sharpen agent i's trade-off in reconciling her adaptation and coordination motives, so that the strategic cost of following more central neighbors outweighs the direct benefit of more precise public information.

Condition (16) also reveals a sharp distributional implication. An agent can be harmed only if she is less central than at least one neighbor she follows. Hence, any agent whose centrality is at least as high as that of all her neighbors, including the most central agents, necessarily benefits from an increase in the precision of public information.[9] Thus, adverse effects arise from relative heterogeneity in network position rather than low centrality per se.

Part (ii) explains when these individual losses translate into a social loss. The aggregate consequence of an agent's distortion depends on how influential her action is for others' coordination. The weighted in-degree index captures this influence by measuring the total coordination weight directed toward a specific agent, and hence the extent to which distortions in her action enter the payoffs of her followers. When the precision of public information increases, a detrimental effect on social welfare requires the coexistence of sufficiently large influence and sufficiently high Katz–Bonacich centrality: formally, $d_i^{\text{in}} > 1$ and $\widehat{b}_i > \frac{d_i^{\text{in}}}{d_i^{\text{in}}-1}$. Neither centrality nor influence alone is sufficient. The social value of public information therefore depends on the extent to which strong responsiveness to public information coincides with strong influence on others.

The mechanism in Murayama (2026) provides a useful exogenous information benchmark: with the precision of private information held fixed, network heterogeneity affects welfare through the extent to which common-noise distortions are transmitted across network positions. Endogenous information acquisition adds a *reinforcing channel*, because a variation in the precision of public information also alters the amount of private information agents acquire. Specifically, the agents most responsive to the public signal reduce acquisition of private information most sharply, so the positions at which common noise is most strongly amplified are also

[9] This observation rules out negative private values of public information when agents have the same aggregate coordination motives, i.e., $\alpha_i = \alpha$ for all $i \in N$. In this case, the equilibrium interaction network $\widehat{\mathbf{G}}$ is regular ($\widehat{\mathbf{G}}\mathbf{1} = \alpha\mathbf{1}$) and has identical Katz–Bonacich centrality across agents.

those where public information crowds out the most private information. Because acquisition is chosen according to private returns, these adjustments need not account for their consequences for followers. The distinction from the exogenous benchmark is therefore fundamental: public information now changes not only how information is used across the network, but also how much private information is produced at each network position. Corollary 2 formalizes this additional margin by comparing the private and social value of public information under exogenous and endogenous private information acquisition.

Corollary 2. *Suppose that the acquisition cost is sufficiently low.*

(i) *The crowding-out effect reduces the private value of public information, with the reduction approximately proportional to the agent's Katz–Bonacich centrality minus one:*

$$\frac{\partial U_i(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{\partial \tau_y} - \frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{d\tau_y} = c(\widehat{b}_i - 1) + O(c^{3/2}), \tag{18}$$

for each agent such that $\alpha_i > 0$.

(ii) *The crowding-out effect reduces the social value of public information, with the reduction approximately proportional to the sum of the products of each agent's Katz–Bonacich centrality and weighted in-degree:*

$$\frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{\partial \tau_y} - \frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{d\tau_y} = c \sum_{i \in N} d_i^{\text{in}} \widehat{b}_i + O(c^{3/2}). \tag{19}$$

Proof: See Appendix A.5. □

Part (i) isolates the channel through which endogenous acquisition diminishes the private value of public information. To understand this result, suppose that other agents' strategies are held fixed. By the envelope theorem, a marginal change in an individual's acquisition choice has no payoff effect at equilibrium.[10] The reduction instead originates from neighboring acquisition responses: as public information becomes more precise, the neighbors being followed cut private acquisition, reducing the role of private information in the actions that matter for coordination. The identity

$$\widehat{b}_i - 1 = \sum_{j \neq i} \widehat{g}_{ij} \widehat{b}_j$$

makes this interpretation explicit: the relevant term excludes agent i's own component and

[10] This envelope argument holds with other agents' strategies fixed. In equilibrium comparative statics, however, a change in an agent's acquisition choice also induces adjustments in other agents' equilibrium actions, which can feed back into her payoff. These indirect effects are of higher order as acquisition costs become small and are absorbed into the $O(c^{3/2})$ remainder.

aggregates the centralities of the neighbors she follows along direct and indirect interaction paths. Hence, stronger links to central neighbors are associated with a larger decline in the private value of public information relative to exogenous information.

Part (ii) gives the corresponding aggregate implication. The social cost of crowding out depends on both where private acquisition falls and how valuable the displaced information is to others. Katz–Bonacich centrality governs the first margin, while weighted in-degree captures the second. Their product captures the contribution of each network position to the reduction in the social value of public information induced by endogenous information acquisition. To make the channel explicit, the total derivative of welfare with respect to τ_y can be written as

$$\frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{d\tau_y} = \frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{\partial \tau_y} + \sum_{i \in N} \frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{\partial \tau_i} \frac{d\widehat{\tau}_i}{d\tau_y}. \tag{20}$$

The first term is the direct effect holding the precisions of private information fixed at their equilibrium levels. The second term captures the additional effect through endogenous information acquisition and is negative: more precise public information reduces the acquisition of private information, while the displaced information retains positive social value at socially influential positions.[11] The welfare loss is amplified when public information displaces private information from socially influential positions. Endogenous acquisition therefore depresses the social value of public information and can overturn an otherwise positive welfare effect, as illustrated by the following example.

Example 1. Consider the directed core–periphery interaction network illustrated in Figure 1, with two core agents, C_1 and C_2, and two peripheral agents, P_1 and P_2. The two core agents follow each other with unit weight, $g_{C_1C_2} = g_{C_2C_1} = 1$, while each peripheral agent follows both core agents with weight one-half, $g_{P_iC_1} = g_{P_iC_2} = \frac{1}{2}$ for $i = 1, 2$. Let α_C and α_P denote the aggregate coordination motives common to core and peripheral agents, respectively. By symmetry, the two core agents share the same equilibrium Katz–Bonacich centrality, denoted by $\widehat{b}_C$, and the two peripheral agents share the same centrality, denoted by $\widehat{b}_P$. They are given by

$$\widehat{b}_C = \frac{1}{1 - \alpha_C}, \qquad \widehat{b}_P = 1 + \frac{\alpha_P}{1 - \alpha_C}.$$

[11] To see this more clearly, Corollary 1 gives $d\widehat{\tau}_i / d\tau_y = -\widehat{b}_i$. With the action rules fixed at their equilibrium values, an increase in agent i's private information acquisition yields a marginal reduction of $\widehat{g}_{ji}\, \widehat{\beta}_i^2 / \widehat{\tau}_i^2 = c\, \widehat{g}_{ji}$ in follower j's expected coordination loss by (10). Summing across followers gives the first-order social benefit $c\, d_i^{\text{in}}$. The induced adjustments in equilibrium actions contribute only higher-order terms, which are absorbed into $O(c^{3/2})$.

The corresponding weighted in-degrees are

$$d_C^{\text{in}} = \alpha_C + \alpha_P, \qquad d_P^{\text{in}} = 0.$$

It is immediate that the core agents are more central than the peripheral agents, $\widehat{b}_C > \widehat{b}_P$, if and only if $\alpha_C > \alpha_P$, whereas the core agents are always more influential in the sense that $d_C^{\text{in}} > d_P^{\text{in}}$.

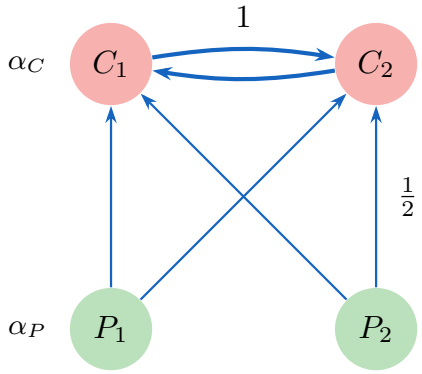


Figure 1: A directed core–periphery interaction network.

Note: Arrows point from followers to the agents they follow. α_C and α_P denote the aggregate coordination motives shared by core and peripheral agents, respectively. Thick arrows indicate weight one between core agents, while thin arrows indicate weight one-half from peripheral agents to core agents.

By Proposition 2, the private value of public information is positive for both core agents. For each peripheral agent, it is negative if and only if

$$(1 - \alpha_C + \alpha_P)^2 < \alpha_C \alpha_P.$$

Thus, a negative private value can arise only for peripheral agents and only when $\alpha_C > \alpha_P$, so that the agents they follow are more central than themselves.

The social value of public information is negative if and only if

$$(1 - \alpha_C + \alpha_P)^2 - \alpha_C \alpha_P < -(1 - \alpha_C^2).$$

The social-loss region is a strict subset of the region in which peripheral agents lose, because their losses must also outweigh the positive gains to the core. The additional term $1 - \alpha_C^2 > 0$ in the above condition captures this requirement.

Figure 2 compares these negative-value regions with the exogenous benchmark, evaluated at the equilibrium precision vector of private information $\widehat{\tau}$.[12] In Panel (a), the additional orange region shows where the acquisition response overturns a positive private value for

[12] Corollary 2 and its proof provide closed-form relationships between the social value of public information under exogenous and endogenous acquisition in the zero-cost limit. These relationships allow us to recover the

peripheral agents. Panel (b) shows the corresponding reversal in social value: the losses to the two peripheral agents exceed the gains to the core once acquisition adjusts.

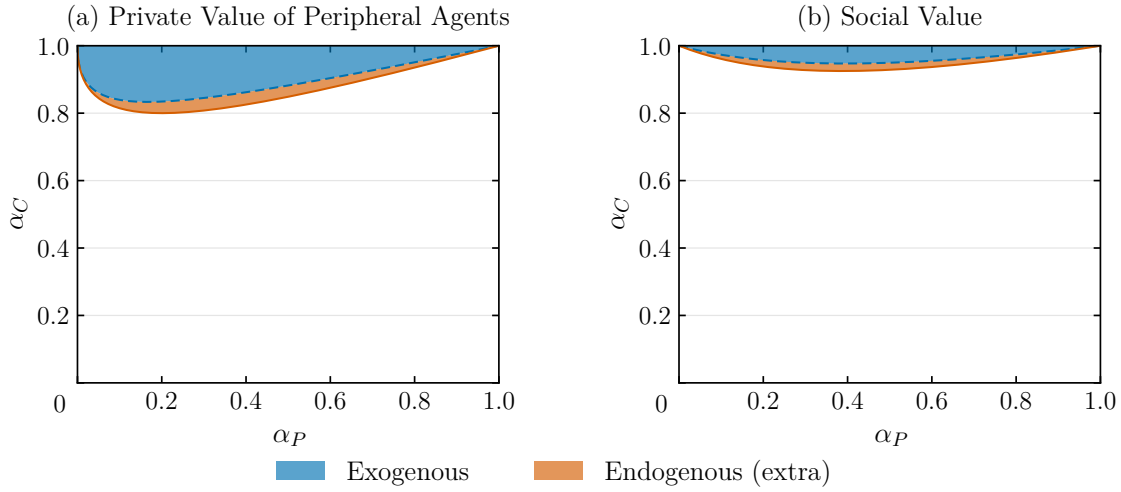


Figure 2: Negative private and social values of public information.

Note: The network contains two core and two peripheral agents. Panels (a) and (b) report negative-value regions for peripheral agents and social welfare, respectively. The blue area denotes the negative-value region under exogenous acquisition, while the orange area denotes the additional cases in which the value becomes negative under endogenous acquisition. The dashed line indicates the boundary of the exogenous region, and the solid line indicates the boundary of the endogenous region. Both regions are evaluated under the low-cost condition.

4 Efficiency and Policy Implications

Following Colombo et al. (2014), we compare the equilibrium with two social optima. Utilitarian welfare is the sum of agents' expected payoffs net of acquisition costs. In both benchmarks, the planner determines the precision of each agent's private information while leaving information dispersed as in Angeletos and Pavan (2007): each agent observes only her own private signal and the public signal, and private information cannot be transferred across agents. We restrict attention to action strategies that are linear in signal realizations. At the social optimum under efficient use of information, the planner also assigns action rules measurable with respect to each agent's own information. We first define this benchmark formally.

Definition 3. A vector of precisions of private information $\boldsymbol{\tau}^* \equiv (\tau_i^*)_{i\in N}$ and a profile of action strategies $\boldsymbol{\sigma}^* \equiv (\sigma_i^*)_{i\in N}$ constitute a socially optimal allocation under efficient use of information if the planner jointly determines the precisions of private information and action

exogenous negative-value regions for sufficiently small acquisition costs. Unlike Murayama (2026), who assumes a common precision of private information across agents, our exogenous benchmark fixes generally heterogeneous precisions at their endogenous equilibrium values.

rules to maximize social welfare:

$$(\boldsymbol{\tau}^*, \boldsymbol{\sigma}^*) \in \underset{\boldsymbol{\tau}\in\mathbb{R}^n_+,\, \boldsymbol{\sigma}\in\Sigma}{\arg\max} \left\{ \sum_{i\in N} \mathbb{E}[u_i(\boldsymbol{\sigma}(\mathbf{x}, y; \boldsymbol{\tau}), \theta; \boldsymbol{\alpha}, \mathbf{G})] - c \sum_{i\in N} \tau_i \right\}.$$

The second benchmark retains decentralized information use. At the social optimum under equilibrium use of information, the planner chooses only the precisions of private information, while agents continue to choose actions according to their equilibrium strategies. For any precision vector $\boldsymbol{\tau}$, let $\boldsymbol{\sigma}^{\text{eq}}(\mathbf{x}, y; \boldsymbol{\tau})$ denote the induced equilibrium profile of action strategies. We define the corresponding allocation as follows.

Definition 4. A vector of precisions of private information $\boldsymbol{\tau}^{**} \equiv (\tau_i^{**})_{i\in N}$ and a profile of action strategies $\boldsymbol{\sigma}^{**} \equiv (\sigma_i^{**})_{i\in N}$ constitute a socially optimal allocation under equilibrium use of information if the planner determines the precisions of private information to maximize social welfare while agents use the resulting information according to the equilibrium action rules:

$$\boldsymbol{\tau}^{**} \in \underset{\boldsymbol{\tau}\in\mathbb{R}^n_+}{\arg\max} \left\{ \sum_{i\in N} \mathbb{E}[u_i(\boldsymbol{\sigma}^{\text{eq}}(\mathbf{x}, y; \boldsymbol{\tau}), \theta; \boldsymbol{\alpha}, \mathbf{G})] - c \sum_{i\in N} \tau_i \right\},$$

with

$$\boldsymbol{\sigma}^{**} \equiv \boldsymbol{\sigma}^{\text{eq}}(\cdot, \cdot; \boldsymbol{\tau}^{**}).$$

Comparing equilibrium with the social optimum under equilibrium use of information isolates the role of information acquisition; comparing the two social optima identifies the welfare consequences of inefficient use of information. These comparisons therefore separate the two sources of inefficiency studied below.

4.1 Efficient Acquisition and Use of Information

The social optimum under efficient use of information. Taking the precisions of private information as given, the planner chooses action rules that satisfy the first-order condition with respect to agent i's action a_i:

$$a_i = \big(1 - \sum_{j\neq i} \widetilde{g}_{ij}\big)\mathbb{E}[\theta \mid x_i, y] + \sum_{j\neq i} \widetilde{g}_{ij}\mathbb{E}[a_j \mid x_i, y], \tag{21}$$

where $\widetilde{g}_{ij}$ captures the social importance the planner assigns to coordinating agent i's action with agent j's. Specifically,

$$\widetilde{g}_{ij} = \frac{\widehat{g}_{ij} + \widehat{g}_{ji}}{1 + d_i^{\text{in}}} \in [0, 1), \tag{22}$$

where d_i^{in} is the weighted in-degree defined in Section 3.2. The numerator in (22) accounts for the coordination relationship between agents i and j in both directions, whereas the denominator captures the total social weight attached to deviations in agent i's action. The planner nevertheless retains a positive adaptation motive toward the fundamental, since

$$1 - \sum_{j \neq i} \widetilde{g}_{ij} = 1 - \frac{\alpha_i + d_i^{\text{in}}}{1 + d_i^{\text{in}}} = \frac{1 - \alpha_i}{1 + d_i^{\text{in}}} > 0.$$

In the following, we refer to $\widetilde{\mathbf{G}} \equiv (\widetilde{g}_{ij})_{i,j \in N}$ as the *efficient interaction network*.

Under efficient use of information, it is readily shown that the weights on the private and public signals also sum to one. [13] We can therefore write the planner's action rule as

$$a_i = \beta_i^* x_i + (1 - \beta_i^*) y.$$

Substituting this rule into (21) yields

$$\beta_i^* = \gamma_i^* \left(1 - \sum_{j \neq i} \widetilde{g}_{ij} + \sum_{j \neq i} \widetilde{g}_{ij} \beta_j^*\right), \tag{23}$$

where τ_i^* is the precision of the private signal acquired by agent i, and $\gamma_i^* \equiv \frac{\tau_i^*}{\tau_i^* + \tau_y}$ is the posterior weight on the private signal. Given the precisions of private information, (23) uniquely determines the efficient use of information.

Turning to information acquisition, we apply the envelope theorem to the planner's problem to obtain

$$c = (1 + d_i^{\text{in}}) \left(\frac{\beta_i^*}{\tau_i^*}\right)^2. \tag{24}$$

Relative to the equilibrium condition in (10), the planner assigns a higher marginal value to the precision of private information because more accurate information benefits not only agent i but also the followers $j \in N_i^{\text{in}}$. The factor $(1 + d_i^{\text{in}})$ captures this additional social value. We postpone the full characterization of this allocation until after introducing the other socially optimal benchmark, which allows the two allocations to be compared directly.

The social optimum under equilibrium use of information. At the social optimum under equilibrium use of information, the planner chooses the precision of each agent's private information, while the mapping from information to actions remains the equilibrium one. Given

[13] Appendix A.6 establishes this property.

the precisions of private information, agent i follows the linear action rule

$$a_i = \beta_i^{**} x_i + (1 - \beta_i^{**}) y,$$

where the weight on private information is determined by the equilibrium mapping in (9):

$$\beta_i^{**} = \mathcal{B}_i^{\text{eq}}(\boldsymbol{\tau}^{**}, \tau_y; \boldsymbol{\alpha}, \mathbf{G}). \tag{25}$$

More precise private information benefits the acquiring agent and her followers. It also changes how agents respond to their information through strategic interactions. The planner accounts for both effects when determining the optimal level of information acquisition. Treating gross welfare $\sum_{k \in N} \mathbb{E}[u_k]$ as a function of $\boldsymbol{\beta}$ and $\boldsymbol{\tau}$, we obtain the interior acquisition condition

$$c = (1 + d_i^{\text{in}}) \left(\frac{\beta_i^{**}}{\tau_i^{**}} \right)^2 + \sum_{j \in N} \frac{\partial \left(\sum_{k \in N} \mathbb{E}[u_k] \right)}{\partial \beta_j} \frac{\partial \mathcal{B}_j^{\text{eq}}}{\partial \tau_i}, \tag{26}$$

where all derivatives are evaluated at the socially optimal allocation $(\boldsymbol{\tau}^{**}, \boldsymbol{\beta}^{**})$, holding $(\boldsymbol{\alpha}, \mathbf{G})$ and τ_y fixed. The first term is the marginal social benefit of more precise private information holding action rules fixed. The second captures the welfare effect of changes in equilibrium rules as agents adjust their reliance on private and public information. Under efficient use of information, the planner's optimal action rules make this second effect vanish by the envelope theorem. Under equilibrium use of information, agents choose action rules according to their private incentives, so the adjustment need not be neutral for social welfare. The planner must therefore account for how information acquired at one position changes information use elsewhere in the network.

We combine the conditions for information acquisition and use to characterize the two socially optimal allocations. To simplify the expressions, define

$$e_i \equiv (1 + d_i^{\text{in}})^{-1/2}, \qquad \mathbf{e} \equiv (e_i)_{i \in N}.$$

Proposition 3. *For each $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$, there exists $\bar{c} > 0$ such that, if the acquisition cost is sufficiently low ($0 < c < \bar{c}$)[14], the two socially optimal allocations are characterized as follows for every agent $i \in N$.*

(i) *At the social optimum under efficient use of information, the weight agent i places on private*

[14]For part (i), a sufficient interiority bound is $c < (\tau_y \max_{i \in N} b_i^*)^{-2}$. For part (ii), the characterization is obtained from a zero-cost expansion and therefore applies for sufficiently small c. The common threshold $\bar{c}$ is chosen so that both parts hold.

information and the precision of private information are given by

$$\beta_i^* = 1 - \tau_y \sqrt{c}\, b_i^* \in (0,1), \tag{27}$$

and

$$\tau_i^* = \sqrt{1 + d_i^{\text{in}}} \left(\frac{1}{\sqrt{c}} - \tau_y b_i^* \right) > 0, \tag{28}$$

where $b_i^* \equiv b_i(\widetilde{\mathbf{G}}, 1, \mathbf{e})$ *is agent i's* **e***-weighted Katz–Bonacich centrality in the efficient interaction network* $\widetilde{\mathbf{G}}$*.*

(ii) *At the social optimum under equilibrium use of information, the weight agent i places on private information and the precision of private information are given by*

$$\beta_i^{**} = 1 - \tau_y \sqrt{c}\, b_i^{**} + O(c) \in (0,1), \tag{29}$$

and

$$\tau_i^{**} = \sqrt{\frac{1 + d_i^{\text{in}}}{c}} + O(1) > 0, \tag{30}$$

where $b_i^{**} \equiv b_i(\widehat{\mathbf{G}}, 1, \mathbf{e})$ *is agent i's* **e***-weighted Katz–Bonacich centrality in the equilibrium interaction network* $\widehat{\mathbf{G}}$*.*

Proof: See Appendix A.8. □

Proposition 3 shows two sources of inefficiency in the equilibrium allocation of information. To understand these results, note that the differences between equilibrium and the two socially optimal benchmarks can be summarized by three Katz–Bonacich centrality vectors:

$$\widehat{\mathbf{b}} = \mathbf{b}(\widehat{\mathbf{G}}, 1, \mathbf{1}), \qquad \mathbf{b}^* = \mathbf{b}(\widetilde{\mathbf{G}}, 1, \mathbf{e}), \qquad \mathbf{b}^{**} = \mathbf{b}(\widehat{\mathbf{G}}, 1, \mathbf{e}).$$

First, both social optima internalize the social benefits of private information to followers, which are reflected in the **e**-weighted Katz–Bonacich centralities b_i^* and b_i^{**}. The weighting vector **e** introduces weighted in-degree into the valuation of information along coordination paths. Second, the social optimum under efficient use of information can also address the inefficiencies in how agents use information through their action rules. The equilibrium interaction network $\widehat{\mathbf{G}}$ is replaced by the efficient interaction network $\widetilde{\mathbf{G}}$, which accounts for coordination relationships in both directions.

We next turn to the comparative statics of socially optimal acquisition with respect to the precision of public information. Under efficient use of information, private acquisition necessarily falls as public information becomes more precise, and the magnitude of this crowding-out

effect has a simple characterization.

Corollary 3. *Suppose that the acquisition cost is sufficiently low. An increase in the precision of public information reduces the precision of private information acquired by each agent at the social optimum under efficient use of information:*

$$\frac{\partial \tau_i^*}{\partial \tau_y} = -\sqrt{1 + d_i^{\text{in}}} b_i^* < 0. \tag{31}$$

Proof. Both d_i^{in} and b_i^* depend only on $(\boldsymbol{\alpha}, \mathbf{G})$. Differentiating (28) with respect to τ_y gives the stated derivative. The Neumann expansion in Definition 2 implies $b_i^* > 0$. □

At the social optimum under efficient use of information, the crowding-out effect of public information also varies systematically across network positions. Unlike in equilibrium, its strength is governed by the efficient centrality b_i^* rather than the equilibrium centrality $\widehat{b}_i$. In addition, the planner values private information from a broader social perspective by internalizing the benefits it confers on followers. Private information is therefore more valuable at influential positions, leading the planner to assign higher precision at these positions and to reduce it more sharply as public information becomes more precise. Crowding out is consequently pronounced at socially influential and central network positions.

However, the crowding-out result does not extend to the social optimum under equilibrium use of information. When the planner controls acquisition but not information use, a more precise public signal can substitute for private information, while also prompting the planner to adjust acquisition to mitigate the distortion arising from equilibrium information use. These two forces can affect acquisition in opposite directions. To obtain more intuition, we provide a numerical example by extending the network in Example 1 to two core agents and m peripheral agents. The core agents continue to follow each other with unit weight, and each peripheral agent follows both core agents with weight one-half; all other link weights are zero.

Figure 3 illustrates the acquisition of private information as the precision of public information varies. In equilibrium and at the social optimum under efficient use of information, acquisition decreases at both network positions. At the social optimum under equilibrium use, acquisition by a peripheral agent also decreases, whereas acquisition by a core agent first rises and then falls. When public information is relatively imprecise, the planner increases acquisition at the core to offset the distortion in equilibrium information use. As public information becomes more precise, the substitution away from private information eventually dominates this corrective response.

The two socially optimal benchmarks differ in how acquisition responds to public information, but both account for the benefits that an agent's private information provides to her

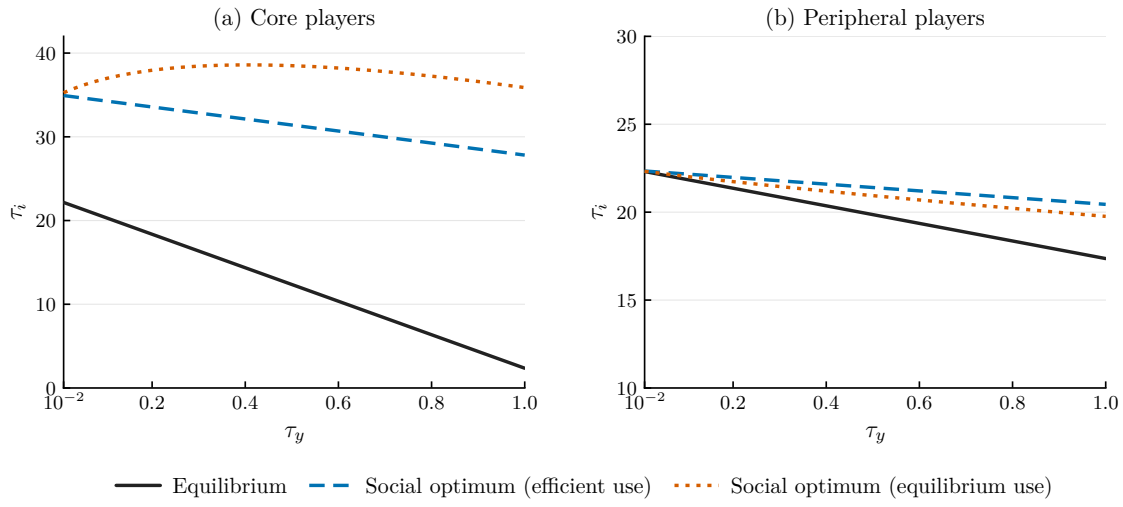


Figure 3: Acquisition of private information in a directed core–periphery network.

Note: The network extends Example 1 to two core agents and $m = 5$ peripheral agents. Panels (a) and (b) report the precision of private information acquired by each core and peripheral agent, respectively. Parameters are $\alpha_C = 0.95$, $\alpha_P = 0.20$, and $c = 2 \times 10^{-3}$, with $\tau_y \in [10^{-2}, 1]$. The curves report exact acquisition choices, all of which are interior.

followers. In equilibrium, by contrast, each agent chooses the precision of private information without internalizing the benefits to others. This discrepancy between private and social returns leads to *underacquisition* at positions with positive social influence, as the following corollary establishes.

Corollary 4. *Suppose that the acquisition cost is sufficiently low. The equilibrium acquisition of private information is inefficiently low,*

$$\widehat{\tau}_i < \tau_i^* \quad \text{and} \quad \widehat{\tau}_i < \tau_i^{**}, \tag{32}$$

for every i such that $d_i^{\text{in}} > 0$. And if the interaction network is non-trivial, the total acquisition of private information in equilibrium is strictly lower than the socially optimal levels:

$$\sum_{i \in N} \widehat{\tau}_i < \sum_{i \in N} \tau_i^*, \quad \text{and} \quad \sum_{i \in N} \widehat{\tau}_i < \sum_{i \in N} \tau_i^{**}. \tag{33}$$

Proof: See Appendix A.9. □

The result is of interest in the sense that it shows the inefficiency of equilibrium information acquisition relative to the social optimum. Notice that the comparison is strict at every position with positive weighted in-degree $d_i^{\text{in}} > 0$; that is, influential agents underinvest in private information in equilibrium. Because the interaction network is nontrivial, at least one such position exists, which also makes the comparison between the aggregate acquisition in equilibrium and the socially optimal levels strict. We also need to emphasize that this result

compares equilibrium separately with the socially optimal benchmarks; it does not rank the social optimum under efficient use of information against the social optimum under equilibrium use of information.

4.2 The Social Value under Socially Optimal Benchmarks

We now ask whether socially optimal acquisition eliminates the detrimental effect of public information documented in Proposition 2.

Proposition 4. *The following results hold with endogenous acquisition of private information.*

(i) *For each* $(\boldsymbol{\alpha}, \mathbf{G})$ *and* $\tau_y > 0$*, there exists* $\bar{c} > 0$ *such that, if the acquisition cost is sufficiently low* $(0 < c < \bar{c})$*, the social value of public information is strictly positive at the social optimum under efficient use of information.*

(ii) *For almost every* $(\boldsymbol{\alpha}, \mathbf{G})$ *and* $\tau_y > 0$*, there exists* $\bar{c} > 0$ *such that, if the acquisition cost is sufficiently low* $(0 < c < \bar{c})$*, the social value of public information is negative at the social optimum under equilibrium use of information if and only if*

$$\sum_{i \in N} \left[(1 - d_i^{\text{in}})(b_i^{**})^2 + \frac{2 d_i^{\text{in}}}{\sqrt{1 + d_i^{\text{in}}}} b_i^{**} \right] < 0. \tag{34}$$

Proof: See Appendix A.10. □

Proposition 4 separates the roles of information acquisition and information use in determining the welfare effect of public information. Part (i) shows that the adverse social value of public information disappears once information acquisition and use are both efficient. At the joint optimum, the planner adjusts both the precision of private information and the action rules optimally to the information environment. By the envelope theorem, the welfare effects operating through these endogenous responses vanish to first order, leaving only the direct informational benefit from an increase in the precision of public information. Accordingly,

$$\frac{dW(\boldsymbol{\tau}^*, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{d\tau_y} = c \sum_{i \in N} \sqrt{1 + d_i^{\text{in}}}\, b_i^* > 0.$$

Network structure therefore affects the magnitude, but not the sign, of the social value of public information under joint efficiency.

Part (ii) demonstrates that efficient acquisition of private information alone is not sufficient to guarantee a positive social value of public information. Socially optimal acquisition corrects the underacquisition of private information at influential positions, but does not alter the private coordination incentives governing information use. As a result, common noise can remain

socially costly when agents at influential positions are also highly responsive to the public signal: errors in the common signal affect their actions, spill over to their followers, and propagate further through the network.

Condition (34) captures this interaction between influence and exposure. Its positive component reflects the benefit of allocating more precise private information to socially important positions, whereas the negative component captures the coordination losses generated by common noise at those same positions. When influence and exposure are sufficiently concentrated, the latter can outweigh the direct informational benefit of more precise public information. Thus, efficient acquisition of information can mitigate, but not eliminate, the welfare distortion arising from equilibrium use of information.

The remaining distortion in information use raises a natural question about how the planner adjusts private information acquisition as public information becomes more precise.

Corollary 5. *Suppose that the acquisition cost is sufficiently low. At the social optimum under equilibrium use of information, the social value of public information is negative if and only if more precise public information increases the total acquisition of private information:*

$$\frac{dW(\boldsymbol{\tau}^{**}, \tau_y; \boldsymbol{\alpha}, \mathbf{G})}{d\tau_y} < 0 \quad \Longleftrightarrow \quad \frac{d}{d\tau_y} \sum_{i \in N} \tau_i^{**} > 0. \tag{35}$$

Proof: See Appendix A.11. □

Corollary 5 reveals a sharp difference between the two social optima. At the social optimum under efficient use of information, an increase in the precision of public information crowds out private information at every position, and thus the total acquisition of private information decreases. At the social optimum under equilibrium use of information, by contrast, the total acquisition of private information can increase with the precision of public information. This *crowding-in* effect arises if and only if the social value of public information is negative. The intuition is that, when public information is socially detrimental, the equilibrium use of information has become excessively dependent on the common signal from the planner's perspective. In response, the planner increases the precision of private information to mitigate such negative effects.

We extend Example 1 to compare the social value of public information at equilibrium with that at the social optimum under equilibrium use of information. The network consists of two core agents who follow each other with unit weight and m peripheral agents who each follow both core agents with weight one-half. Figure 4 depicts the negative-value regions in the (α_P, α_C) plane, based on the low-cost conditions in (17) and (34). In each panel, public information has

negative social value at equilibrium above the dashed boundary and under socially optimal acquisition within the shaded region. In this example, the latter region lies strictly inside the former.[15] For parameter combinations between the dashed boundary and the lower edge of the shaded region, correcting the acquisition distortion reverses the social value from negative to positive. Within the shaded region, however, public information remains socially detrimental even when private information acquisition is socially optimal.

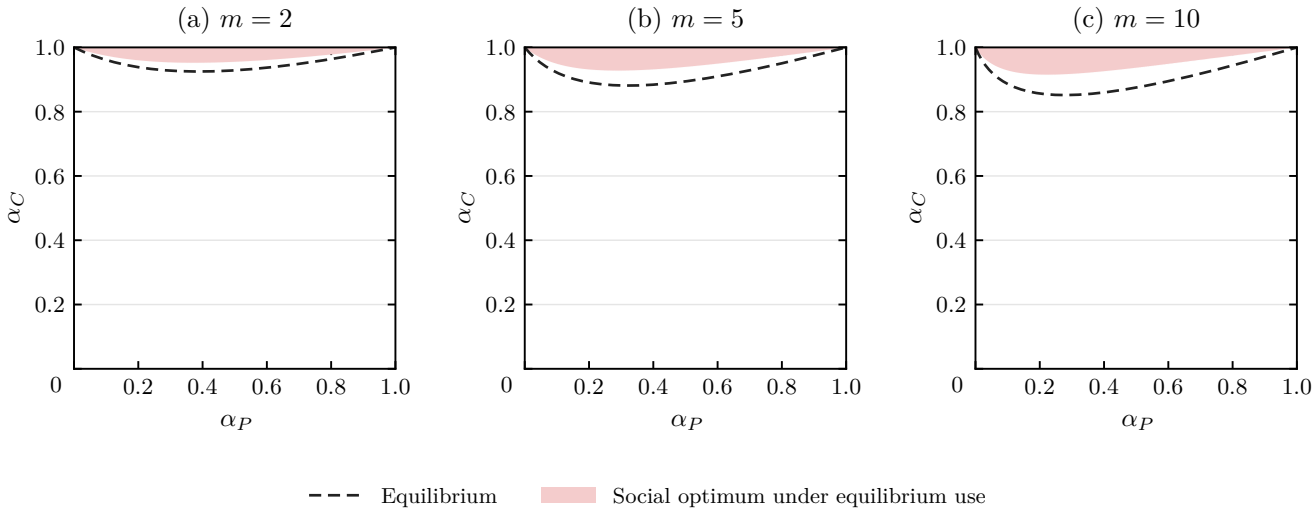


Figure 4: Negative social value of public information in a directed core–periphery network.

Note: The network extends Example 1 to two core agents and m peripheral agents, with $m = 2$, 5, and 10 in panels (a), (b), and (c), respectively. The region above the dashed boundary has a negative social value in equilibrium; the shaded region indicates a negative social value at the social optimum under equilibrium use of information. Both regions are evaluated under the low-cost condition.

4.3 Policy Interventions

The preceding welfare analysis shows that restoring efficiency may require addressing both private information acquisition and information use. This distinction naturally motivates the policy interventions considered below. Pigouvian taxes and subsidies are standard instruments for correcting inefficiencies in decentralized equilibria. Taxes make decision makers internalize the costs their choices impose on others, whereas subsidies reward activities that generate external benefits. With dispersed information, however, corrective policies must also account for how information is acquired and subsequently used, as emphasized by Angeletos and Pavan (2009) and Colombo et al. (2026). In our network setting, these two dimensions give rise to distinct externalities: an agent's action affects her neighbors' coordination losses, while more

[15] The nesting shown in this example is not a general property: the social value of public information is not systematically ordered across the two allocations. For instance, in a regular network, the social value is always positive at equilibrium but can be negative when information acquisition is efficient subject to equilibrium information use.

precise private information also benefits those who follow her. These externalities suggest two corresponding interventions: a Pigouvian tax on realized actions and a subsidy for private information acquisition.

For either intervention, the planner commits to the tax or subsidy schedule before Stage I. The schedule and the model primitives are common knowledge. Tax payments and subsidy receipts enter agents' payoffs additively, all payments are enforceable, and agents face no liquidity constraints. Acquisition choices and deviations remain unobservable to other agents. Welfare is the sum of agents' expected payoffs less the total cost of information acquisition, evaluated at the acquisition choices and actions induced by the policy. Tax and subsidy payments cancel when agents' payoffs are consolidated with the planner's budget.

Both instruments include a fixed transfer $\ell_i \in \mathbb{R}$ for each agent $i \in N$, where $\ell_i > 0$ denotes a payment by the agent to the planner and $\ell_i < 0$ denotes a payment to the agent. These transfers are fixed before Stage I and do not depend on subsequent actions or acquisition choices. The role of the transfer scheme is to balance the planner's budget in expectation at the induced equilibrium.[16]

Taxes on information use. We first consider taxes on information use. We assume that the planner can perfectly observe agents' realized actions when taxes are collected after Stage II. One interpretation is a research organization that adjusts analysts' compensation ex post based on discrepancies among their forecasts to align private incentives with organizational objectives.

For each agent $i \in N$, the planner specifies a twice continuously differentiable tax schedule $t_i(\mathbf{a})$ contingent on the realized action profile $\mathbf{a}$. As in Angeletos and Pavan (2009), we require the tax schedule to preserve the quadratic structure of payoffs. Agent i's net payment to the planner is

$$\mathcal{T}_i(\mathbf{a}) = t_i(\mathbf{a}) + \ell_i. \tag{36}$$

Agent i's payoff net of acquisition costs and transfers is

$$u_i(\mathbf{a}, \theta; \boldsymbol{\alpha}, \mathbf{G}) - c\tau_i - t_i(\mathbf{a}) - \ell_i. \tag{37}$$

Since the tax is contingent on realized actions, agent i takes its expected marginal cost into account when deciding how to use her information. The optimal action at Stage II satisfies

$$a_i = (1 - \alpha_i)\mathbb{E}[\theta \mid x_i, y] + \sum_{j \neq i} \widehat{g}_{ij}\mathbb{E}[a_j \mid x_i, y] - \frac{1}{2}\mathbb{E}\Big[\frac{\partial t_i(\mathbf{a})}{\partial a_i} \mid x_i, y\Big]. \tag{38}$$

[16] When the intervention yields a strictly positive welfare gain, fixed transfers can distribute this gain so that every agent benefits strictly ex ante relative to laissez faire. Appendix A.15 provides the construction.

The expected marginal tax therefore directly affects information use at Stage II and, in turn, the incentive to acquire private information at Stage I. A change in the precision of private information affects both the informativeness of the subsequent action and the expected tax payment generated by that action. At an interior acquisition choice with linear information use, the envelope theorem gives[17]

$$c = \frac{\beta_i^2}{\tau_i^2}\Big[1 + \frac{1}{2}\mathbb{E}\big(\frac{\partial^2 t_i(\mathbf{a})}{\partial a_i^2}\big)\Big]. \tag{39}$$

The next proposition constructs a Pigouvian tax that aligns private and social incentives in the action stage and thereby implements the social optimum under efficient information use.

Proposition 5. *For each $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$, there exists $\bar{c} > 0$ such that, if the acquisition cost is sufficiently low ($0 < c < \bar{c}$), the socially optimal allocation under efficient use of information $(\boldsymbol{\tau}^*, \boldsymbol{\beta}^*)$ can be implemented as the unique linear equilibrium by setting each agent's tax payment to*

$$t_i(\mathbf{a}) = \sum_{j \neq i} \alpha_j g_{ji}(a_i - a_j)^2. \tag{40}$$

Proof: See Appendix A.12. □

The Pigouvian tax internalizes the coordination losses that agent i's action imposes on her followers. Without intervention, agent i considers only the coordination losses with her leaders, while the tax makes her also bear the losses imposed on her followers. For a *pure* follower, that is, an agent with $d_i^{\text{in}} = 0$, the tax is zero. For an influencer, that is, an agent with $d_i^{\text{in}} > 0$, define the weighted average action of followers as

$$\bar{a}_i^F \equiv \frac{\sum_{j \neq i} \alpha_j g_{ji} a_j}{d_i^{\text{in}}}.$$

The marginal tax is then

$$\frac{\partial t_i(\mathbf{a})}{\partial a_i} = 2 d_i^{\text{in}}(a_i - \bar{a}_i^F).$$

At Stage II, agent i evaluates the marginal tax conditional on (x_i, y), creating an incentive to move a_i toward $\mathbb{E}[\bar{a}_i^F \mid x_i, y]$. A larger weighted in-degree strengthens this incentive. The tax makes an agent's influence on followers enter the action decision.

Importantly, the same tax also changes the return to private information acquisition. Since

$$\frac{1}{2}\frac{\partial^2 t_i(\mathbf{a})}{\partial a_i^2} = \sum_{j \neq i} \alpha_j g_{ji} = d_i^{\text{in}},$$

[17] This first-order condition holds under the regularity conditions stated in Appendix A.12, which are satisfied by the quadratic tax considered below.

the acquisition condition in (39) coincides with the planner's envelope condition under efficient use of information in (24). Thus, although the tax is levied only on realized actions, it implements the jointly efficient allocation by correcting incentives for both information acquisition and use.

Subsidies on information acquisition. We next consider subsidies for information acquisition. We assume that the planner can perfectly observe the precision of private information acquired by each agent at the time subsidies are paid. One interpretation is a research organization that subsidizes analysts' information purchases based on confidential acquisition records that identify the purchased data products and their known quality.

For each agent $i \in N$, the planner chooses a subsidy rate $s_i \in [0, 1)$. Agent i's net payment to the planner is

$$\mathcal{T}_i(\tau_i) = \ell_i - c s_i \tau_i, \tag{41}$$

so her payoff net of acquisition costs and transfers is

$$u_i(\mathbf{a}, \theta; \boldsymbol{\alpha}, \mathbf{G}) - c(1 - s_i)\tau_i - \ell_i. \tag{42}$$

Because the transfer depends only on the acquisition choice, and not on actions or signal realizations, it does not alter the strategic incentives underlying subsequent information use. Hence, conditional on the precisions of private information $\boldsymbol{\tau}$, agents' optimal actions at Stage II remain characterized by the equilibrium mapping in (9).

The information acquisition problem at Stage I differs from its laissez-faire counterpart only through the subsidized acquisition cost $c(1 - s_i)$. Applying the same envelope argument yields the following condition for an interior acquisition choice:

$$c(1 - s_i) = \left(\frac{\beta_i}{\tau_i}\right)^2. \tag{43}$$

Our next result documents that the planner can implement the social optimum under equilibrium use of information by equating each agent's subsidized marginal cost of information acquisition to her private marginal benefit at that allocation.

Proposition 6. *For each $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$, there exists $\bar{c} > 0$ such that, if the acquisition cost is sufficiently low ($0 < c < \bar{c}$), the socially optimal allocation under equilibrium use of information $(\boldsymbol{\tau}^{**}, \boldsymbol{\beta}^{**})$ can be implemented as the unique linear equilibrium by setting each agent's subsidy rate to*

$$s_i^{**} = 1 - \frac{1}{c}\left(\frac{\beta_i^{**}}{\tau_i^{**}}\right)^2 = \frac{d_i^{\text{in}}}{1 + d_i^{\text{in}}} + O(\sqrt{c}), \tag{44}$$

where β_i^{**} and τ_i^{**} are the weight agent i places on private information and the acquired precision of private information, respectively. The implementing subsidy rates satisfy $s_i^{**} \in [0,1)$, with $s_i^{**} > 0$ if and only if $d_i^{\text{in}} > 0$.

Proof: See Appendix A.13. □

The subsidy on information acquisition is *targeted* across agents according to their positions in the interaction network. Its first-order component internalizes the external benefit of more precise private information to followers, as summarized by weighted in-degree. The exact implementing rate further accounts for the endogenous adjustment in equilibrium information use induced by changes in private acquisition.

For a sufficiently small acquisition cost, the expansion in Proposition 6 admits a *simple subsidy rule* for each agent $i \in N$:

$$s_i^0 = \frac{d_i^{\text{in}}}{1 + d_i^{\text{in}}}. \tag{45}$$

This rule targets subsidies according to social influence. Intuitively, improving the precision of private information acquired by an agent with a larger weighted in-degree generates greater aggregate benefits for followers, so more influential positions warrant stronger incentives for information acquisition.

Let $\mathbf{s} = (s_i)_{i \in N}$ denote the vector of subsidy rates. The corresponding equilibrium is described by the precision vector $\boldsymbol{\tau}(\mathbf{s})$ and the vector of coefficients on private information $\boldsymbol{\beta}(\mathbf{s})$. The next corollary characterizes the equilibrium under the simple subsidy rule and shows that the social value of public information can remain negative even as social welfare improves.

Corollary 6. *Suppose that the acquisition cost is sufficiently low. At the equilibrium induced by the simple subsidy rule $\mathbf{s}^0 \equiv (s_i^0)_{i \in N}$, the weight agent i places on private information and the precision of private information are*

$$\beta_i(\mathbf{s}^0) = 1 - \tau_y \sqrt{c}\, b_i^{**} \in (0,1), \tag{46}$$

$$\tau_i(\mathbf{s}^0) = \sqrt{1 + d_i^{\text{in}}} \left(\frac{1}{\sqrt{c}} - \tau_y b_i^{**} \right) > 0. \tag{47}$$

Social welfare necessarily increases relative to the laissez-faire equilibrium:

$$W(\boldsymbol{\tau}(\mathbf{s}^0), \tau_y; \boldsymbol{\alpha}, \mathbf{G}) - W(\widehat{\boldsymbol{\tau}}, \tau_y; \boldsymbol{\alpha}, \mathbf{G}) = \sqrt{c} \sum_{i \in N} \left(\sqrt{1 + d_i^{\text{in}}} - 1 \right)^2 + O(c) > 0, \tag{48}$$

while the social value of public information is negative if and only if condition (34) holds.

Proof: See Appendix A.14. □

The welfare gain from the simple subsidy rule reflects the external value of private information at influential network positions. However, this improvement does not imply that more precise public information must be socially beneficial. Indeed, the condition for a negative social value coincides with that at the social optimum under equilibrium use of information.

Discussion. The two policy instruments provide an implementation counterpart to the two socially optimal benchmarks. A subsidy for private information acquisition corrects the acquisition externality while deliberately leaving information use decentralized, and therefore implements the social optimum subject to equilibrium use of information. An action tax operates on a different margin: by incorporating the social incentives into individual actions, it directly corrects information use and, through the induced continuation value, also affects information acquisition. The action tax requires realized actions to be observable and bilateral interaction strengths to be known, whereas the subsidy requires both verifiable acquisition of private information and information about the network structure. The simple subsidy rule is less demanding in this respect, as it depends only on weighted in-degree. Which instrument is more appropriate in practice therefore depends in part on what the planner can observe, verify, and contract upon.

A second implication concerns the interpretation of transparency policy. Improving the incentives to produce private information need not make public information socially desirable. For instance, the simple subsidy rule raises welfare relative to laissez faire, yet the social value of public information can remain negative because information use remains governed by private coordination incentives. Hence, the level of welfare under an intervention and the marginal value of public disclosure are conceptually distinct. A policy that improves decentralized information acquisition may therefore coexist with a continued case for caution in expanding public disclosure.

5 Conclusion

This paper investigates the welfare effects of public information when agents endogenously acquire private information and interact through a network. We show that endogenous information acquisition reduces the private and social value of public information and can overturn a positive effect present in the exogenous information setting. Comparing equilibrium with two socially optimal benchmarks reveals systematic underacquisition of private information. Public information is necessarily welfare improving when both information acquisition and use are efficient, but can remain socially detrimental when only acquisition is efficient. We further characterize two policy interventions that implement the corresponding social optima.

These findings have a broader implication for transparency policy in environments with dispersed information and heterogeneous interactions. Regulators and firms often seek to improve decision making by releasing more precise public information, yet disclosure can also reshape private information production and the strategic use of information by market participants. Its welfare effects therefore depend on the surrounding incentive and interaction structure—including the rewards to proprietary research, the extent of benchmarking, and the influence of different participants. Transparency policy should accordingly be evaluated not only by the accuracy of the information released, but also by how disclosure alters the information environment in which decisions are made.

Future research could examine how the social value of public information changes when the interaction network is endogenous, only partially known, or composed of multiple layers. Allowing agents to choose their coordination partners would link information acquisition to network formation. With incomplete knowledge of the network, information acquisition and use would also depend on beliefs about others' interactions. Multiplex networks would allow agents to make related decisions across different layers of relationships. These extensions would clarify how the welfare effects of public information depend on the formation, observability, and organization of the networks in which agents interact.

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Online Appendix for

"Information Acquisition and the Value of Public Information in Network Games"

Jiacheng Fan

Lingnan College, Sun Yat-sen University

fanjch7@mail2.sysu.edu.cn

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Appendix A. Proofs of the Main Results

A.1 The Coefficients in the Linear Equilibrium Rule

Consider the linear action rule

$$a_i = \widehat{w}_i^x x_i + \widehat{w}_i^y y. \tag{A.1}$$

Lemma 1. *In a linear equilibrium, the coefficients in* (A.1) *sum to one for every* $i \in N$*:*

$$\widehat{w}_i^x + \widehat{w}_i^y = 1. \tag{A.2}$$

Proof. We show that the sums of the coefficients solve a linear system whose unique solution is the vector of ones. Let $\widehat{w}_i \equiv \widehat{w}_i^x + \widehat{w}_i^y$ and $\widehat{\mathbf{w}} \equiv (\widehat{w}_i)_{i\in N}$. Independence of the signal errors gives

$$\mathbb{E}[a_j \mid x_i, y] = \widehat{w}_j^x \mathbb{E}[\theta \mid x_i, y] + \widehat{w}_j^y y, \qquad j \neq i.$$

The coefficients on x_i and y in the posterior mean of θ sum to one. Substituting the preceding expectation into (6) and adding the coefficients on these two signals therefore yields

$$\widehat{w}_i = (1 - \alpha_i) + \sum_{j\neq i} \widehat{g}_{ij} \widehat{w}_j.$$

In vector form, this is

$$(\mathbf{I} - \widehat{\mathbf{G}})\widehat{\mathbf{w}} = \mathbf{1} - \boldsymbol{\alpha}.$$

The row sums of $\widehat{\mathbf{G}}$ imply $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{1} = \mathbf{1} - \boldsymbol{\alpha}$. Moreover, $\rho(\widehat{\mathbf{G}}) < 1$ makes the matrix invertible. Thus $\widehat{\mathbf{w}} = \mathbf{1}$. If an agent acquires no private information, her rule depends only on y and her coefficient on private information is zero. Matching the coefficient on y gives the same system. The conclusion therefore also covers zero acquisition. □

A.2 Proof of Proposition 1

We first characterize information use for a given precision vector. We then solve the acquisition problem under unobservable deviations and establish uniqueness, including at the boundary.

Information use for given precisions. Lemma 1 allows us to write each linear equilibrium rule as $a_i = \beta_i x_i + (1 - \beta_i) y$. For positive τ_i, matching the coefficient on x_i in (6) gives

$$\frac{\beta_i}{\tau_i} = \frac{(1 - \alpha_i) - \beta_i + \sum_{j\neq i} \widehat{g}_{ij} \beta_j}{\tau_y}. \tag{A.3}$$

This condition is sufficient for sequential optimality: conditional expected payoff is strictly concave in a_i, with second derivative -2. Independence of the signal errors also gives the expected adaptation and coordination losses:

$$\mathbb{E}(a_i - \theta)^2 = \frac{\beta_i^2}{\tau_i} + \frac{(1-\beta_i)^2}{\tau_y}, \tag{A.4}$$

$$\mathbb{E}(a_i - a_j)^2 = \frac{\beta_i^2}{\tau_i} + \frac{\beta_j^2}{\tau_j} + \frac{(\beta_i - \beta_j)^2}{\tau_y}. \tag{A.5}$$

Differentiating expected payoff with respect to β_i gives (A.3) as well, because $\sum_{j\neq i} \widehat{g}_{ij} = \alpha_i$. The second derivative with respect to β_i is $-2(1/\tau_i + 1/\tau_y) < 0$.

Rearranging (A.3) yields (8). Let $\boldsymbol{\gamma} \equiv (\gamma_i(\tau_i))_{i\in N}$. The row sums satisfy

$$\rho\left(\operatorname{diag}(\boldsymbol{\gamma})\widehat{\mathbf{G}}\right) \leq \max_{i\in N} \gamma_i(\tau_i)\alpha_i < 1.$$

The system thus has a unique solution within the class of linear rules. The inverse of its coefficient matrix is nonnegative, so $\boldsymbol{\beta} \geq \mathbf{0}$. In addition,

$$\mathbf{1} - \boldsymbol{\beta} = \left[\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}\right]^{-1} (\mathbf{1} - \boldsymbol{\gamma}) > \mathbf{0}.$$

Hence $0 \leq \beta_i < 1$, with $\beta_i = 0$ when $\tau_i = 0$ and $\beta_i > 0$ when $\tau_i > 0$.

Unobservable acquisition deviations. Consider a deviation to τ_i' by agent i. Other agents retain their equilibrium action rules because they do not observe the deviation. Substituting these rules into the action condition gives the deviating agent's optimal coefficient:

$$\beta_i(\tau_i') = \frac{\tau_i'}{\tau_i' + \tau_y}\left[1 - \alpha_i + \sum_{j\neq i} \widehat{g}_{ij}\widehat{\beta}_j\right].$$

The term in brackets is fixed during the deviation and is strictly positive because $1 - \alpha_i > 0$ and the remaining terms are nonnegative. Denote this term by q_i. The envelope theorem eliminates the change in her own optimal coefficient, as in the information-acquisition argument of Colombo et al. (2014, Appendix A, proof of Proposition 1), so her marginal net payoff from acquisition is

$$\frac{q_i^2}{(\tau_i' + \tau_y)^2} - c.$$

Its derivative with respect to τ_i' is $-2q_i^2/(\tau_i' + \tau_y)^3 < 0$. The acquisition problem consequently has a unique maximizer, equal to $\max\{0, q_i/\sqrt{c} - \tau_y\}$. At an interior equilibrium, the first-order

condition becomes

$$\widehat{\tau}_i = \frac{\widehat{\beta}_i}{\sqrt{c}}.$$

Substituting this relation into (8) and canceling $\widehat{\beta}_i > 0$ gives

$$\widehat{\beta}_i - \sum_{j \neq i} \widehat{g}_{ij} \widehat{\beta}_j = 1 - \alpha_i - \tau_y \sqrt{c}.$$

Since $(\mathbf{I} - \widehat{\mathbf{G}})\mathbf{1} = \mathbf{1} - \boldsymbol{\alpha}$ and $(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{1} = \widehat{\mathbf{b}}$, the interior candidate is

$$\widehat{\beta}_i = 1 - \tau_y \sqrt{c}\, \widehat{b}_i, \qquad \widehat{\tau}_i = \frac{1}{\sqrt{c}} - \tau_y \widehat{b}_i.$$

Uniqueness and interiority. The acquisition best response, including its zero-precision boundary, and sequentially optimal use together imply

$$\beta_i = \max \left\{ 0,\, 1 - \alpha_i - \tau_y \sqrt{c} + \sum_{j \neq i} \widehat{g}_{ij} \beta_j \right\}. \tag{A.6}$$

Taking the positive part does not increase distances. The map in (A.6) is therefore a contraction in the supremum norm with modulus at most $\alpha_{\max} \equiv \max_{i \in N} \alpha_i < 1$, and it has a unique fixed point. For $0 < c < (\tau_y \max_i \widehat{b}_i)^{-2}$, the interior candidate satisfies $0 < \widehat{\beta}_i < 1$ and $\widehat{\tau}_i > 0$ for every agent. It is consequently the unique fixed point. The associated action rules are sequentially optimal, and each acquisition choice is a global best response to the other agents' rules. This establishes the unique linear equilibrium and the formulas in (12) and (13). □

A.3 Welfare Identities and Responses of Information Use

We suppress the fixed arguments $\boldsymbol{\alpha}$ and $\mathbf{G}$ in payoffs, while retaining the precision arguments $(\boldsymbol{\tau}, \tau_y)$ in $U_i(\boldsymbol{\tau}, \tau_y)$ and $W(\boldsymbol{\tau}, \tau_y)$. Each payoff is evaluated under the action rules specified in the calculation.

To derive total expected loss, fix positive precisions and normalized linear rules $a_i = \beta_i x_i + (1 - \beta_i) y$, treating the precisions and coefficients as independent choices. Welfare under these rules is the sum of expected primitive payoffs net of acquisition costs. Substituting the adaptation

loss from (A.4) and the coordination loss from (A.5) into that sum gives

$$\begin{aligned}
-W(\boldsymbol{\tau}, \tau_y) &= -\sum_{i \in N} \mathbb{E}[u_i(\mathbf{a}, \theta)] + c \sum_{i \in N} \tau_i \\
&= \sum_{i \in N} (1 - \alpha_i) \left[\frac{\beta_i^2}{\tau_i} + \frac{(1 - \beta_i)^2}{\tau_y} \right] \\
&\quad + \sum_{i \in N} \sum_{j \neq i} \widehat{g}_{ij} \left[\frac{\beta_i^2}{\tau_i} + \frac{\beta_j^2}{\tau_j} + \frac{(\beta_i - \beta_j)^2}{\tau_y} \right] + c \sum_{i \in N} \tau_i \\
&= \sum_{i \in N} (1 + d_i^{\text{in}}) \frac{\beta_i^2}{\tau_i} + c \sum_{i \in N} \tau_i \\
&\quad + \frac{1}{\tau_y} \left[\sum_{i \in N} (1 - \alpha_i)(1 - \beta_i)^2 + \sum_{i \in N} \sum_{j \neq i} \widehat{g}_{ij} (\beta_i - \beta_j)^2 \right].
\end{aligned}$$

In the last equality, the private signal error of agent i enters her own expected loss with coefficient one and her followers' expected losses with aggregate coefficient d_i^{in}. This gives the coefficient $1 + d_i^{\text{in}}$ on β_i^2 / τ_i.

The expression in square brackets measures the total loss attributable to the common public signal error, before division by its precision. It depends on each agent's coefficient on the public signal. Write this coefficient as $\eta_i \equiv 1 - \beta_i$ and collect the coefficients in $\boldsymbol{\eta} \equiv (\eta_i)_{i \in N} = \mathbf{1} - \boldsymbol{\beta}$. We denote the resulting quadratic form on $\mathbb{R}^n$ by

$$\mathcal{L}(\boldsymbol{\eta}) \equiv \sum_{i \in N} (1 - \alpha_i) \eta_i^2 + \sum_{i \in N} \sum_{j \neq i} \widehat{g}_{ij} (\eta_i - \eta_j)^2. \tag{A.7}$$

Total expected loss can therefore be written as

$$-W(\boldsymbol{\tau}, \tau_y) = \sum_{i \in N} (1 + d_i^{\text{in}}) \frac{(1 - \eta_i)^2}{\tau_i} + \frac{\mathcal{L}(\boldsymbol{\eta})}{\tau_y} + c \sum_{i \in N} \tau_i. \tag{A.8}$$

The welfare in (A.8) is evaluated at the stated coefficients, before imposing any restriction on information use. Under equilibrium use, substitute $\boldsymbol{\beta} = \mathcal{B}^{\text{eq}}(\boldsymbol{\tau}, \tau_y)$, where we also suppress the fixed network arguments in the equilibrium mapping. Under efficient use, evaluate the same expression at the planner's coefficients.

We next derive the response of equilibrium information use to a change in the precision vector, holding τ_y fixed. The notation $\boldsymbol{\beta}(\boldsymbol{\tau})$ in the following system denotes the equilibrium mapping at this fixed τ_y. Thus the entire profile of action rules adjusts to each candidate precision

vector, and its coefficients satisfy

$$\boldsymbol{\beta}(\boldsymbol{\tau}) = \boldsymbol{\Gamma}(\boldsymbol{\tau})\left[\mathbf{1} - \boldsymbol{\alpha} + \widehat{\mathbf{G}}\boldsymbol{\beta}(\boldsymbol{\tau})\right]. \tag{A.9}$$

Only the ith diagonal entry of $\boldsymbol{\Gamma}$ changes directly with τ_i. Differentiating the system, moving the induced changes in neighbors' coefficients to the left, and inverting gives, for every $k \in N$,

$$\frac{\partial \beta_k(\boldsymbol{\tau})}{\partial \tau_i} = \left(\left[\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau})\widehat{\mathbf{G}}\right]^{-1}\right)_{ki} \gamma_i'(\tau_i)\left[1 - \alpha_i + \left(\widehat{\mathbf{G}}\boldsymbol{\beta}(\boldsymbol{\tau})\right)_i\right], \tag{A.10}$$

where $\gamma_i'(\tau_i) = \tau_y/(\tau_i + \tau_y)^2 > 0$. The inverse captures the feedback through equilibrium action choices.

A.4 Proof of Proposition 2

Private value. We first evaluate individual welfare at the interior acquisition equilibrium and then differentiate it along the equilibrium precision vector $\widehat{\boldsymbol{\tau}}(\tau_y)$. For any positive precision vector, substituting the equilibrium action rules into the primitive payoff gives

$$\begin{aligned} -U_i(\boldsymbol{\tau}, \tau_y) = \frac{\beta_i^2}{\tau_i} + \sum_{j \neq i} \widehat{g}_{ij}\frac{\beta_j^2}{\tau_j} + c\tau_i \\ + \frac{(1-\alpha_i)(1-\beta_i)^2 + \sum_{j\neq i}\widehat{g}_{ij}(\beta_i - \beta_j)^2}{\tau_y}, \end{aligned} \tag{A.11}$$

where $\boldsymbol{\beta} = \mathcal{B}^{\text{eq}}(\boldsymbol{\tau}, \tau_y)$. At the interior acquisition equilibrium, $\widehat{\beta}_i = 1 - \tau_y\sqrt{c}\,\widehat{b}_i$ and $\widehat{\tau}_i = \widehat{\beta}_i/\sqrt{c}$. Substituting both relations into (A.11) gives the equilibrium expected payoff, including acquisition costs:

$$\begin{aligned} U_i(\widehat{\boldsymbol{\tau}}, \tau_y) = -(2+\alpha_i)\sqrt{c} + c\tau_y\left(2\widehat{b}_i + \sum_{j\neq i}\widehat{g}_{ij}\widehat{b}_j\right) \\ - c\tau_y\left[(1-\alpha_i)\widehat{b}_i^2 + \sum_{j\neq i}\widehat{g}_{ij}(\widehat{b}_i - \widehat{b}_j)^2\right]. \end{aligned}$$

Expanding the square and using $\sum_{j\neq i}\widehat{g}_{ij} = \alpha_i$ and $\sum_{j\neq i}\widehat{g}_{ij}\widehat{b}_j = \widehat{b}_i - 1$ reduces this expression to

$$U_i(\widehat{\boldsymbol{\tau}}, \tau_y) = -(2+\alpha_i)\sqrt{c} + c\tau_y\left[\widehat{b}_i^2 + \widehat{b}_i - 1 - \sum_{j\neq i}\widehat{g}_{ij}\widehat{b}_j^2\right]. \tag{A.12}$$

The centralities depend only on the fixed network. Differentiating along the equilibrium allocation and applying the centrality identity gives

$$\frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} = c\left[\widehat{b}_i^2 - \sum_{j\neq i} \widehat{g}_{ij}\widehat{b}_j(\widehat{b}_j - 1)\right].$$

Since $c > 0$ and $\widehat{g}_{ij} = \alpha_i g_{ij}$, the sign comparison is

$$\frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} \gtrless 0 \quad \Longleftrightarrow \quad \widehat{b}_i^2 \gtrless \alpha_i \sum_{j\neq i} g_{ij}\widehat{b}_j(\widehat{b}_j - 1).$$

The negative case is (16), which proves part (i).

For the distributional implication, suppose $\widehat{b}_i \geq \widehat{b}_j$ for every j with $\widehat{g}_{ij} > 0$. Every centrality is at least one, so

$$\sum_{j\neq i} \widehat{g}_{ij}\widehat{b}_j(\widehat{b}_j - 1) \leq (\widehat{b}_i - 1)\sum_{j\neq i} \widehat{g}_{ij}\widehat{b}_j = (\widehat{b}_i - 1)^2.$$

It follows that

$$\frac{dU_i(\widehat{\boldsymbol{\tau}}(\tau_y), \tau_y)}{d\tau_y} \geq c(2\widehat{b}_i - 1) > 0.$$

Social value. Sum (A.12) over agents. Regrouping the network term by the agent being followed gives $\sum_i \sum_{j\neq i} \widehat{g}_{ij}\widehat{b}_j^2 = \sum_j d_j^{\text{in}}\widehat{b}_j^2$. Equilibrium welfare is therefore

$$W(\widehat{\boldsymbol{\tau}}, \tau_y) = c\tau_y \sum_{i\in N}\left[(1 - d_i^{\text{in}})\widehat{b}_i^2 + \widehat{b}_i - 1\right] - \sqrt{c}\left(2n + \sum_{i\in N}\alpha_i\right). \tag{A.13}$$

Differentiating along the equilibrium allocation and using $\sum_i(\widehat{b}_i - 1) = \sum_i d_i^{\text{in}}\widehat{b}_i$ yields

$$\begin{aligned}\frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} &= c\sum_{i\in N}\left[(1 - d_i^{\text{in}})\widehat{b}_i^2 + \widehat{b}_i - 1\right] \\ &= c\sum_{i\in N}\left[(1 - d_i^{\text{in}})\widehat{b}_i^2 + d_i^{\text{in}}\widehat{b}_i\right].\end{aligned}$$

Because $c > 0$,

$$\frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} \gtrless 0 \quad \Longleftrightarrow \quad \sum_{i\in N}\left[(1 - d_i^{\text{in}})\widehat{b}_i^2 + d_i^{\text{in}}\widehat{b}_i\right] \gtrless 0.$$

The negative case is (17), which proves part (ii). □

A.5 Proof of Corollary 2

Fix the network and $\tau_y > 0$ and let $c \to 0$. Partial derivatives of U_i and W hold the other precision arguments fixed while allowing equilibrium information use to adjust; they are then evaluated at $\boldsymbol{\tau} = \widehat{\boldsymbol{\tau}}$. Total derivatives with respect to τ_y also include the acquisition response $d\widehat{\tau}_k/d\tau_y = -\widehat{b}_k$. In both comparisons, we calculate the partial derivative minus the total derivative.

First, we record a bound on the induced changes in information use that applies to both the private and social comparisons. The row sums imply

$$\left\| \left[\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau}) \widehat{\mathbf{G}} \right]^{-1} \right\|_{\infty} \leq \frac{1}{1 - \alpha_{\max}}, \qquad \alpha_{\max} = \max_{i \in N} \alpha_i < 1.$$

At the equilibrium precision vector, $\widehat{\tau}_k = 1/\sqrt{c} - \tau_y \widehat{b}_k$, so

$$\gamma_k'(\widehat{\tau}_k) = \frac{\tau_y}{(\widehat{\tau}_k + \tau_y)^2} = O(c).$$

The remaining factor in (A.10) is bounded because $0 < \widehat{\beta}_j < 1$. Hence, for all $j, k \in N$,

$$\left. \frac{\partial \beta_j}{\partial \tau_k} \right|_{\boldsymbol{\tau} = \widehat{\boldsymbol{\tau}}} = O(c).$$

Private value. To compute the precision derivatives of $U_i(\boldsymbol{\tau}, \tau_y)$, first return to the primitive expression on the right-hand side of (A.11), before substituting the equilibrium mapping for $\boldsymbol{\beta}$. Its negative is individual expected payoff net of acquisition cost with the coefficients and precisions treated as independent variables. Holding all coefficients fixed, the derivative of expected payoff net of acquisition costs with respect to τ_i is zero at the equilibrium allocation, since $\widehat{\beta}_i^2/\widehat{\tau}_i^2 = c$. For $k \neq i$, the derivative with respect to τ_k is $c\widehat{g}_{ik}$. These are the direct acquisition effects.

The derivatives with respect to coefficients below also refer to this expression for expected payoff net of acquisition costs before the equilibrium mapping is substituted. Its derivative with respect to agent i's own coefficient is zero at equilibrium by her action first-order condition. For a neighbor's coefficient, the derivative at the equilibrium allocation is

$$-2\widehat{g}_{ij} \left[\frac{\widehat{\beta}_j}{\widehat{\tau}_j} + \frac{\widehat{\beta}_j - \widehat{\beta}_i}{\tau_y} \right] = -2\sqrt{c}\, \widehat{g}_{ij}(1 + \widehat{b}_i - \widehat{b}_j), \qquad j \neq i.$$

Each such derivative is $O(\sqrt{c})$. Multiplying it by the $O(c)$ response of information use and summing over finitely many agents contributes $O(c^{3/2})$. Restoring the induced changes in use

therefore gives

$$\left.\frac{\partial U_i(\boldsymbol{\tau}, \tau_y)}{\partial \tau_k}\right|_{\boldsymbol{\tau}=\widehat{\boldsymbol{\tau}}} = c\widehat{g}_{ik} + O(c^{3/2}),$$

including $k = i$, because $\widehat{g}_{ii} = 0$. The full derivative with respect to own precision can thus contain a term of order $c^{3/2}$ even though its direct acquisition component vanishes.

The chain rule now yields

$$\begin{aligned}\frac{\partial U_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} - \frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} &= \sum_k \widehat{b}_k \frac{\partial U_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_k} \\ &= c \sum_{j \neq i} \widehat{g}_{ij}\widehat{b}_j + O(c^{3/2}) \\ &= c(\widehat{b}_i - 1) + O(c^{3/2}).\end{aligned}$$

This proves (18). When $\alpha_i > 0$, the nonnegative network and the centrality identity imply $\widehat{b}_i > 1$, so the leading term is strictly positive. A common sufficiently small cost bound applies to all such agents because the population is finite. If $\alpha_i = 0$, agent i's action and payoff are independent of every other agent's precision, and her own acquisition derivative is exactly zero at her interior optimum. Her acquisition contribution is therefore zero throughout the interior region.

Social value. We apply the same chain rule to welfare. First differentiate the negative of the primitive total loss in (A.8), holding $\boldsymbol{\eta}$ fixed. The direct effect, net of acquisition cost and evaluated at the equilibrium allocation, is

$$(1 + d_i^{\text{in}})\frac{\widehat{\beta}_i^2}{\widehat{\tau}_i^2} - c = c\, d_i^{\text{in}} = c \sum_{j \neq i} \widehat{g}_{ji}.$$

The equality uses the private acquisition condition. It leaves the benefit of agent i's information to her followers.

The induced change in information use contributes only a higher-order term. At the equilibrium allocation, $\boldsymbol{\eta} = \mathbf{1} - \widehat{\boldsymbol{\beta}} = O(\sqrt{c})$ and $1/\widehat{\tau}_k = O(\sqrt{c})$. Since $\mathcal{L}$ is quadratic, differentiating the primitive loss in (A.8) with respect to η_k at fixed precisions gives

$$\frac{\partial}{\partial \eta_k}\left[\sum_{\ell \in N}(1 + d_\ell^{\text{in}})\frac{(1-\eta_\ell)^2}{\tau_\ell} + \frac{\mathcal{L}(\boldsymbol{\eta})}{\tau_y}\right] = O(\sqrt{c})$$

at this allocation. The bound established at the start of the proof gives $\partial \eta_k / \partial \tau_i = -\partial \beta_k / \partial \tau_i =$

$O(c)$. The total contribution through information use is consequently $O(c^{3/2}) = o(c)$. Thus

$$\left.\frac{\partial W(\boldsymbol{\tau}, \tau_y)}{\partial \tau_i}\right|_{\boldsymbol{\tau}=\widehat{\boldsymbol{\tau}}} = c\, d_i^{\text{in}} + O(c^{3/2}),$$

so the chain rule gives

$$\begin{aligned}\frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} - \frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} &= \sum_{i \in N} \widehat{b}_i \frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_i} \\ &= c \sum_{i \in N} d_i^{\text{in}} \widehat{b}_i + O(c^{3/2}),\end{aligned}$$

which is (19). Every weighted in-degree is nonnegative, and at least one is positive because the interaction network is nontrivial. All centralities are positive and fixed as $c \to 0$. The leading coefficient $\sum_i d_i^{\text{in}} \widehat{b}_i$ is therefore strictly positive, while the remainder divided by c vanishes. Endogenous acquisition consequently reduces the social value of public information for all sufficiently low acquisition costs. □

A.6 The Coefficients under Efficient Information Use

We hold $(\boldsymbol{\alpha}, \mathbf{G})$ fixed unless stated otherwise and suppress these arguments in welfare expressions. Each such expression is evaluated under the action rule specified for the corresponding allocation.

For a fixed vector of precisions of private information, consider the planner's linear action rule

$$a_i = \widetilde{w}_i^x x_i + \widetilde{w}_i^y y, \tag{A.14}$$

Lemma 2. *At the social optimum under efficient use of information, the coefficients in* (A.14) *sum to one for every $i \in N$:*

$$\widetilde{w}_i^x + \widetilde{w}_i^y = 1. \tag{A.15}$$

Proof.

Let $w_i \equiv \widetilde{w}_i^x + \widetilde{w}_i^y$ denote the sum of agent i's two coefficients, and let $\mathbf{w} \equiv (w_i)_{i \in N}$. Independence of the signal errors implies

$$\mathbb{E}[a_j \mid x_i, y] = \widetilde{w}_j^x \mathbb{E}[\theta \mid x_i, y] + \widetilde{w}_j^y y.$$

The posterior weights on x_i and y sum to one. Substituting this conditional expectation into the

planner's first-order condition (21) and adding its two signal coefficients therefore gives

$$w_i = 1 - \sum_{j \neq i} \widetilde{g}_{ij} + \sum_{j \neq i} \widetilde{g}_{ij} w_j.$$

Equivalently,

$$(\mathbf{I} - \widetilde{\mathbf{G}})(\mathbf{w} - \mathbf{1}) = \mathbf{0}.$$

The efficient interaction network is nonnegative, and its ith row sums to $(\alpha_i + d_i^{\text{in}})/(1 + d_i^{\text{in}}) < 1$. Thus $\rho(\widetilde{\mathbf{G}}) < 1$, which makes $\mathbf{I} - \widetilde{\mathbf{G}}$ invertible. The preceding system has the unique solution $\mathbf{w} = \mathbf{1}$, as claimed. □

A.7 A Low-Cost Lemma for Equilibrium Use of Information

The next lemma establishes a unique optimum when the planner chooses acquisition and agents retain the equilibrium action rules $a_i = \beta_i(\boldsymbol{\tau}) x_i + [1 - \beta_i(\boldsymbol{\tau})] y$. It also supplies the regularity needed to differentiate welfare at that optimum.

Lemma 3 (Low-cost regularity under equilibrium use)**.** *Fix $\tau_y > 0$. For all sufficiently low c, the social optimum under equilibrium use of information has a unique globally optimal precision vector $\boldsymbol{\tau}^{**}$. For this vector, define*

$$r_i = \frac{1}{\sqrt{c}\,(\tau_i^{**} + \tau_y)}, \qquad \mathbf{r} \equiv (r_i)_{i \in N}.$$

There is a real-analytic map $\bar{\mathbf{r}}$ in a neighborhood of zero such that

$$\mathbf{r} = \bar{\mathbf{r}}(\tau_y \sqrt{c}), \qquad \bar{\mathbf{r}}(0) = \mathbf{e}.$$

Moreover,

$$\mathbf{1} - \boldsymbol{\beta}^{**} = \tau_y \sqrt{c}\,(\mathbf{I} - \widehat{\mathbf{G}})^{-1} \mathbf{e} + O(c). \tag{A.16}$$

Finally, the minimized total loss, divided by $\sqrt{c}$, is represented by a real-analytic scalar function $F(\cdot)$ in a neighborhood of zero. Welfare at this optimum, evaluated under $a_i = \beta_i^{**} x_i + (1 - \beta_i^{**}) y$, satisfies

$$W(\boldsymbol{\tau}^{**}, \tau_y) = -\sqrt{c}\, F(\tau_y \sqrt{c}). \tag{A.17}$$

Proof. Fix the network and $\tau_y > 0$, and write $s \equiv \tau_y \sqrt{c}$. We first identify the limiting trade-off between information losses and acquisition costs, then show that every global optimum approaches its unique solution.

The limiting planner's problem. For any candidate precision vector, measure the posterior variance relative to $\sqrt{c}$:

$$r_i = \frac{1}{\sqrt{c}\,(\tau_i + \tau_y)}, \qquad \tau_i = \frac{1}{\sqrt{c}\,r_i} - \tau_y.$$

This is a one-to-one transformation from $\tau_i \geq 0$ to $0 < r_i \leq 1/s$. Let $H_c(\mathbf{r})$ denote the planner's total loss divided by $\sqrt{c}$. We will show that, for each fixed $\mathbf{r} > \mathbf{0}$,

$$\lim_{c \to 0} H_c(\mathbf{r}) = \sum_i \left[(1 + d_i^{\text{in}}) r_i + \frac{1}{r_i} \right].$$

In this limiting problem, reducing the posterior variance lowers the loss due to private signal errors but raises acquisition costs. Their balance gives the unique minimizer $r_i = e_i = (1 + d_i^{\text{in}})^{-1/2}$. This is the allocation to which we will show that all global optima converge.

Equilibrium use determines the coefficients on the public signal as

$$\boldsymbol{\eta}(\mathbf{r}, c) = \left[\mathbf{I} - \big(\mathbf{I} - s\,\text{diag}(\mathbf{r})\big)\widehat{\mathbf{G}} \right]^{-1} s\mathbf{r}. \tag{A.18}$$

On the feasible set, the matrix $(\mathbf{I} - s\,\text{diag}(\mathbf{r}))\widehat{\mathbf{G}}$ is nonnegative with row sums at most $\alpha_{\max} < 1$, so its resolvent exists and is nonnegative. On every fixed compact subset of the strictly positive orthant, the resolvent is analytic in s near zero and gives

$$\boldsymbol{\eta}(\mathbf{r}, c) = s(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{r} + O(c), \tag{A.19}$$

uniformly together with its first derivatives in $\mathbf{r}$.

Write $\boldsymbol{\eta} = \boldsymbol{\eta}(\mathbf{r}, c)$. Equilibrium use implies $1 - \eta_i = (1 - sr_i)[1 - (\widehat{\mathbf{G}}\boldsymbol{\eta})_i]$, so the normalized loss due to agent i's private signal error is

$$\frac{(1 + d_i^{\text{in}})(1 - \eta_i)^2}{\tau_i \sqrt{c}} = (1 + d_i^{\text{in}}) r_i (1 - sr_i)[1 - (\widehat{\mathbf{G}}\boldsymbol{\eta})_i]^2.$$

The right-hand side also defines its continuous value at $\tau_i = 0$. Since normalized acquisition cost is $c\tau_i/\sqrt{c} = 1/r_i - s$, substitution into (A.8) gives the normalized total loss

$$H_c(\mathbf{r}) = \sum_i (1 + d_i^{\text{in}}) r_i (1 - sr_i)[1 - (\widehat{\mathbf{G}}\boldsymbol{\eta})_i]^2 + \frac{\mathcal{L}(\boldsymbol{\eta})}{s} + \sum_i \left(\frac{1}{r_i} - s \right). \tag{A.20}$$

The network and τ_y are suppressed in this notation. The three terms are the losses due to private signal errors, the loss due to the public signal error, and acquisition costs. Equation (A.19) now gives the limiting objective above.

Locating all global optima. All three terms in (A.20) are nonnegative. For fixed positive c, the feasible set is bounded above, the objective is continuous through its upper faces, and acquisition costs diverge as any $r_i \to 0$. A global minimizer therefore exists. For small $s > 0$, the feasible vector $\mathbf{e}$ supplies a uniform bound $H_c(\mathbf{e}) \leq K$. At every global minimizer, this bound implies $r_i \geq 1/(K+s)$ and $\mathcal{L}(\boldsymbol{\eta}) \leq Ks$. The inequality $\mathcal{L}(\boldsymbol{\eta}) \geq \min_i(1-\alpha_i)\|\boldsymbol{\eta}\|_2^2$ then gives $\boldsymbol{\eta} \to \mathbf{0}$ uniformly over all global minimizers. The nonnegative resolvent in (A.18) also gives $s\mathbf{r} \leq \boldsymbol{\eta}$, hence $s\mathbf{r} \to \mathbf{0}$.

For sufficiently small s, both $1 - sr_i$ and $1 - (\widehat{\mathbf{G}}\boldsymbol{\eta})_i$ are at least $1/2$ at every minimizer. The corresponding term due to the private signal error is therefore at least $(1 + d_i^{\text{in}})r_i/8$, implying $r_i \leq 8K/(1 + d_i^{\text{in}})$. These bounds place every global minimizer in a common compact box in the strictly positive orthant and, for small s, strictly below the upper boundary $1/s$.

The resolvent expansion gives, uniformly on this box,

$$\frac{\partial H_c(\mathbf{r})}{\partial r_i} = 1 + d_i^{\text{in}} - \frac{1}{r_i^2} + O(\sqrt{c}). \tag{A.21}$$

Since every minimizer is interior, every accumulation point satisfies $1 + d_i^{\text{in}} - r_i^{-2} = 0$. The unique positive solution is $\mathbf{e}$, so compactness implies that all global minimizers converge to $\mathbf{e}$.

Uniqueness and analytic welfare. Equation (A.18) expresses $\boldsymbol{\eta}$ as s times an analytic vector. Since $\mathcal{L}$ is quadratic, the apparent singularity in $\mathcal{L}(\boldsymbol{\eta})/s$ is removable. Thus the objective and its first-order system extend analytically near $(s, \mathbf{r}) = (0, \mathbf{e})$. The system's Jacobian in $\mathbf{r}$ is $2\,\mathrm{diag}(e_i^{-3})$ there and is invertible. The analytic implicit-function theorem (Krantz and Parks, 2013, Section 6.1) gives a unique stationary branch $\bar{\mathbf{r}}(s)$ with $\bar{\mathbf{r}}(0) = \mathbf{e}$. Every global minimizer eventually lies in this neighborhood; the branch is therefore the unique global optimum for sufficiently small positive s.

Analyticity gives $\bar{\mathbf{r}}(s) = \mathbf{e} + O(\sqrt{c})$, and (A.19) then proves (A.16). Define the minimized normalized loss by

$$F(s) = H_c(\bar{\mathbf{r}}(s)), \qquad s = \tau_y\sqrt{c} > 0.$$

The objective and feasible set depend on c and τ_y only through s, so the same F applies to nearby positive values of both parameters with the network fixed. The analytic objective and branch extend F to a neighborhood of zero; negative arguments serve only this extension. Welfare under the equilibrium action rules is $-\sqrt{c}$ times the minimized normalized loss, proving (A.17). □

A.8 Proof of Proposition 3

The social optimum under efficient use of information. Within the maintained class of linear action strategies, the planner assigns rules $a_i = \beta_i^* x_i + (1-\beta_i^*)y$. For fixed precisions, a variation in agent i's action affects her own loss and the coordination losses of her followers. Conditioning on (x_i, y) yields (21); substituting the linear rules gives (23). The loss in (A.8) is strictly convex in the public coefficients $\boldsymbol{\eta}$, so these use conditions identify the unique optimum for given precisions.

To optimize acquisition jointly with use, first fix η_i. Minimizing $(1+d_i^{\text{in}})(1-\eta_i)^2/\tau_i + c\tau_i$ over $\tau_i \geq 0$ gives $\tau_i = \sqrt{1+d_i^{\text{in}}}|1-\eta_i|/\sqrt{c}$. Substituting back leaves the reduced loss

$$2\sqrt{c}\sum_i \sqrt{1+d_i^{\text{in}}}\,|1-\eta_i| + \frac{\mathcal{L}(\boldsymbol{\eta})}{\tau_y}. \tag{A.22}$$

This objective is strictly convex: the absolute-value term is convex and $\mathcal{L}$ is positive definite. Its quadratic gradient is

$$\frac{1}{2}\frac{\partial \mathcal{L}(\boldsymbol{\eta})}{\partial \eta_i} = (1+d_i^{\text{in}})\eta_i - \sum_{j\neq i}(\widehat{g}_{ij}+\widehat{g}_{ji})\eta_j.$$

Thus, on the region $\eta_i < 1$ for every i, the first-order conditions reduce to

$$\eta_i - \sum_{j\neq i}\widetilde{g}_{ij}\eta_j = \frac{\tau_y\sqrt{c}}{\sqrt{1+d_i^{\text{in}}}}.$$

The unique solution is $\eta_i^* = \tau_y\sqrt{c}\,b_i^*$. For sufficiently low c, it lies in $(0,1)^n$ and satisfies the first-order conditions of the full convex problem, making it the unique global minimizer. Using $\beta_i^* = 1-\eta_i^*$ gives

$$\beta_i^* = 1-\tau_y\sqrt{c}\,b_i^*, \qquad \tau_i^* = \sqrt{1+d_i^{\text{in}}}\left(\frac{1}{\sqrt{c}} - \tau_y b_i^*\right).$$

Both formulas satisfy the interiority conditions in part (i).

The social optimum under equilibrium use of information. The planner now anticipates $a_i = \beta_i(\boldsymbol{\tau})x_i + [1-\beta_i(\boldsymbol{\tau})]y$. Holding these coefficients fixed, the marginal gross welfare benefit of τ_i in (A.8) is $(1+d_i^{\text{in}})[\beta_i(\boldsymbol{\tau})/\tau_i]^2$. Adding the changes in coefficients from (A.10) and equating the total benefit to c gives (26). Under efficient use, the social gradient in coefficients vanishes; here the planner also accounts for their induced adjustment.

At the optimum, Lemma 3 gives $r_i = e_i + O(\sqrt{c})$. Since $e_i > 0$, taking reciprocals preserves

this order, and

$$\tau_i^{**} = \frac{1}{\sqrt{c}\, r_i} - \tau_y = \frac{1}{\sqrt{c}\, e_i} + O(1) = \sqrt{\frac{1 + d_i^{\text{in}}}{c}} + O(1).$$

Equation (A.16) and $\mathbf{b}^{**} = (\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{e}$ also give

$$\beta_i^{**} = 1 - \tau_y \sqrt{c}\, b_i^{**} + O(c).$$

The positive leading precision and $b_i^{**} > 0$ imply $\tau_i^{**} > 0$ and $0 < \beta_i^{**} < 1$ for sufficiently low c. These are the expansions and interiority claims in part (ii). □

A.9 Proof of Corollary 4

Propositions 1 and 3 give

$$\lim_{c \to 0} \sqrt{c}\, \widehat{\tau}_i = 1, \qquad \lim_{c \to 0} \sqrt{c}\, \tau_i^{*} = \lim_{c \to 0} \sqrt{c}\, \tau_i^{**} = \sqrt{1 + d_i^{\text{in}}}.$$

For $d_i^{\text{in}} > 0$, the limit for each socially optimal precision strictly exceeds the equilibrium limit. Convergence gives the individual comparisons in (32). Summing the limiting gaps gives, for each social optimum,

$$\sum_i \left(\sqrt{1 + d_i^{\text{in}}} - 1 \right) > 0.$$

The strict inequality follows from $\sum_i d_i^{\text{in}} = \sum_i \alpha_i > 0$. It implies the two aggregate comparisons in (33). A common positive cost bound applies to the finitely many individual and aggregate comparisons. □

A.10 Proof of Proposition 4

Fix the network and $\tau_y > 0$. In each case, we differentiate optimized welfare with respect to τ_y at fixed c, within the interior region.

The social optimum under efficient use of information. Evaluate $W(\boldsymbol{\tau}^*, \tau_y)$ under $a_i = \beta_i^* x_i + (1 - \beta_i^*) y$, and write $\boldsymbol{\eta}^* = \mathbf{1} - \boldsymbol{\beta}^*$. The losses due to private signal errors and acquisition costs are equal at this optimum, with sum $2\sqrt{c} \sum_i \sqrt{1 + d_i^{\text{in}}}(1 - \eta_i^*)$. The reduced first-order conditions give $\partial \mathcal{L}(\boldsymbol{\eta}^*)/\partial \eta_i = 2\tau_y \sqrt{c} \sqrt{1 + d_i^{\text{in}}}$. Multiplying the first-order conditions by η_i^* and summing, we use the quadratic identity $\sum_i \eta_i^* \partial \mathcal{L}(\boldsymbol{\eta}^*)/\partial \eta_i = 2\mathcal{L}(\boldsymbol{\eta}^*)$ to obtain

$$\frac{\mathcal{L}(\boldsymbol{\eta}^*)}{\tau_y} = \sqrt{c} \sum_i \sqrt{1 + d_i^{\text{in}}}\, \eta_i^*.$$

Combining the loss components and using $\eta_i^* = \tau_y \sqrt{c}\, b_i^*$ gives

$$W(\boldsymbol{\tau}^*, \tau_y) = -2\sqrt{c} \sum_i \sqrt{1 + d_i^{\text{in}}} + c\tau_y \sum_i \sqrt{1 + d_i^{\text{in}}}\, b_i^*. \tag{A.23}$$

Since centralities are fixed and $b_i^* \geq e_i = (1 + d_i^{\text{in}})^{-1/2}$ by the nonnegative Neumann expansion,

$$\frac{dW(\boldsymbol{\tau}^*, \tau_y)}{d\tau_y} = c \sum_i \sqrt{1 + d_i^{\text{in}}}\, b_i^* \geq cn > 0.$$

Choosing the cost bound within the interior region proves part (i).

The social optimum under equilibrium use of information. Evaluate $W(\boldsymbol{\tau}^{**}, \tau_y)$ under $a_i = \beta_i^{**} x_i + (1 - \beta_i^{**})y$, and put $s = \tau_y \sqrt{c}$. Lemma 3 gives $W(\boldsymbol{\tau}^{**}, \tau_y) = -\sqrt{c} F(s)$, so the social value is $-cF'(s)$. To obtain the sign criterion, we compute $F'(0)$.

At $s = 0$, the gradient of the normalized objective vanishes at $\mathbf{e}$. The envelope theorem therefore eliminates the derivative of the optimal branch and gives

$$H_c(\mathbf{e}) = 2 \sum_i \sqrt{1 + d_i^{\text{in}}} + sF'(0) + O(c). \tag{A.24}$$

To compute this coefficient, substitute $\boldsymbol{\eta}(\mathbf{e}, c) = s\mathbf{b}^{**} + O(c)$ into (A.20). The terms $1 - se_i$ and $1/r_i - s$ together contribute $-2n$ to the derivative, and the squared term contributes $-2\sum_i \sqrt{1 + d_i^{\text{in}}} (\widehat{\mathbf{G}}\mathbf{b}^{**})_i$. Thus

$$\begin{aligned} F'(0) &= \mathcal{L}(\mathbf{b}^{**}) - 2n - 2 \sum_i \sqrt{1 + d_i^{\text{in}}} (\widehat{\mathbf{G}}\mathbf{b}^{**})_i \\ &= \mathcal{L}(\mathbf{b}^{**}) - 2 \sum_i \sqrt{1 + d_i^{\text{in}}}\, b_i^{**}, \end{aligned} \tag{A.25}$$

where the second equality uses $\widehat{\mathbf{G}}\mathbf{b}^{**} = \mathbf{b}^{**} - \mathbf{e}$ and $\sqrt{1 + d_i^{\text{in}}} e_i = 1$. Continuity of F' yields

$$\lim_{c \to 0} \frac{1}{c} \frac{dW(\boldsymbol{\tau}^{**}, \tau_y)}{d\tau_y} = 2 \sum_i \sqrt{1 + d_i^{\text{in}}}\, b_i^{**} - \mathcal{L}(\mathbf{b}^{**}). \tag{A.26}$$

To express this limit in the primitives of the proposition, expand the quadratic form:

$$\begin{aligned} \mathcal{L}(\mathbf{b}^{**}) &= \sum_i (1 + d_i^{\text{in}})(b_i^{**})^2 - 2 \sum_i b_i^{**} (\widehat{\mathbf{G}}\mathbf{b}^{**})_i \\ &= \sum_i \left[(d_i^{\text{in}} - 1)(b_i^{**})^2 + 2e_i b_i^{**} \right]. \end{aligned}$$

The second equality uses the same centrality identity. Substituting into (A.26) and recalling $e_i = (1 + d_i^{\text{in}})^{-1/2}$ gives

$$\lim_{c \to 0} \frac{1}{c} \frac{dW(\boldsymbol{\tau}^{**}, \tau_y)}{d\tau_y} = \sum_i \left[(1 - d_i^{\text{in}})(b_i^{**})^2 + \frac{2d_i^{\text{in}}}{\sqrt{1 + d_i^{\text{in}}}} b_i^{**} \right].$$

Denote this statistic by $\Phi(\boldsymbol{\alpha}, \mathbf{G})$ within this proof. If $\Phi \neq 0$, convergence places the derivative divided by c within $|\Phi|/2$ of Φ for all sufficiently low c. Its sign therefore equals that of Φ.

For the almost-every qualification, consider the relative interior of any admissible parameter face specified by the zero entries of $\boldsymbol{\alpha}$ and $\mathbf{G}$. This set is connected, and Φ is real analytic there: each d_i^{in} is polynomial in the primitives, $\mathbf{e}$ is analytic, and $\rho(\widehat{\mathbf{G}}) < 1$ ensures an analytic resolvent. The function is not identically zero. Holding $\mathbf{G}$ fixed and replacing $\boldsymbol{\alpha}$ by $t\boldsymbol{\alpha}$ for $0 < t \leq 1$ preserves the face and $\alpha_i < 1$. As $t \to 0$, $\widehat{\mathbf{G}} \to \mathbf{0}$, every $d_i^{\text{in}} \to 0$, and both $\mathbf{e}$ and $\mathbf{b}^{**}$ converge to $\mathbf{1}$. Hence $\Phi(t\boldsymbol{\alpha}, \mathbf{G}) \to n > 0$. Applying the zero-set theorem for nontrivial real-analytic functions in affine coordinates on each parameter face (Mityagin, 2020), we conclude that its zero set has relative Lebesgue measure zero on that face. The preceding sign argument proves part (ii). □

A.11 Proof of Corollary 5

Total acquisition is the negative derivative of optimized welfare with respect to its unit cost. To apply this relationship, hold the network and τ_y fixed and evaluate welfare under the equilibrium action rules $a_i = \beta_i^{**} x_i + (1 - \beta_i^{**}) y$. The feasible set of precisions is independent of c, which enters welfare through $-c \sum_i \tau_i$. Lemma 3 supplies a differentiable optimal branch for positive c in the stated range. The envelope theorem (Milgrom and Segal, 2002, Theorem 1) and the lemma's analytic representation $W(\boldsymbol{\tau}^{**}, \tau_y) = -\sqrt{c} F(\tau_y \sqrt{c})$ therefore give

$$\sum_i \tau_i^{**} = -\frac{dW(\boldsymbol{\tau}^{**}, \tau_y)}{dc} = \frac{F(\tau_y \sqrt{c})}{2\sqrt{c}} + \frac{\tau_y}{2} F'(\tau_y \sqrt{c}).$$

Differentiating optimized welfare and this identity with respect to τ_y at fixed c gives

$$\frac{dW(\boldsymbol{\tau}^{**}, \tau_y)}{d\tau_y} = -cF'(\tau_y \sqrt{c}), \tag{A.27}$$

$$\frac{d}{d\tau_y} \sum_i \tau_i^{**} = F'(\tau_y \sqrt{c}) + \frac{\tau_y \sqrt{c}}{2} F''(\tau_y \sqrt{c}). \tag{A.28}$$

Set $s = \tau_y \sqrt{c}$. If F' is not identically zero near zero, let $a \neq 0$ be its first nonzero Taylor

coefficient, of order $m \in \{0, 1, 2, \ldots\}$. Analyticity permits termwise differentiation and gives

$$F'(s) = as^m + O(s^{m+1}), \qquad F'(s) + \frac{s}{2}F''(s) = \left(1 + \frac{m}{2}\right) as^m + O(s^{m+1}).$$

Since $1 + m/2 > 0$, these expressions have the same sign for small positive s. If F' is identically zero, both vanish. Equations (A.27) and (A.28) therefore imply

$$\frac{dW(\boldsymbol{\tau}^{**}, \tau_y)}{d\tau_y} < 0 \quad \Longleftrightarrow \quad \frac{d}{d\tau_y} \sum_i \tau_i^{**} > 0$$

for all sufficiently low c, including cases in which $F'(0) = 0$. □

A.12 Proof of Proposition 5

The tax in (40) makes each agent bear the coordination losses that her action imposes on her followers. Following the policy approach of Angeletos and Pavan (2009), we verify its incentives for both information use and acquisition. The proof establishes efficient actions, globally optimal acquisition following unobservable deviations, and implementation as the unique linear equilibrium.

Efficient actions. Substituting (36) and (40) into agent i's payoff gives

$$u_i - \mathcal{T}_i(\mathbf{a}) = -(1 - \alpha_i)(a_i - \theta)^2 - \sum_{j \neq i} (\widehat{g}_{ij} + \widehat{g}_{ji})(a_i - a_j)^2 - \ell_i. \tag{A.29}$$

Because the conditional objective has second derivative $-2(1 + d_i^{\text{in}}) < 0$, its first-order condition identifies the unique optimal action:

$$(1 + d_i^{\text{in}})a_i = (1 - \alpha_i)\mathbb{E}[\theta \mid x_i, y] + \sum_{j \neq i} (\widehat{g}_{ij} + \widehat{g}_{ji})\mathbb{E}[a_j \mid x_i, y]. \tag{A.30}$$

Dividing by $1 + d_i^{\text{in}}$ yields the efficient information-use condition (21), because

$$\widetilde{g}_{ij} = \frac{\widehat{g}_{ij} + \widehat{g}_{ji}}{1 + d_i^{\text{in}}}, \qquad 1 - \sum_{j \neq i} \widetilde{g}_{ij} = \frac{1 - \alpha_i}{1 + d_i^{\text{in}}} > 0.$$

The positive adaptation motive keeps every row sum of $\widetilde{\mathbf{G}}$ strictly below one. Because (A.30) is the same conditional first-order condition used in Lemma 2, that lemma's coefficient argument applies: every linear solution has coefficients summing to one and can be written as $a_j = \beta_j x_j + (1 - \beta_j)y$. At zero precision, the coefficient on private information is zero under the

maintained convention. For fixed precisions, the coefficient system has matrix $\mathbf{I} - \boldsymbol{\Gamma}(\boldsymbol{\tau})\widetilde{\mathbf{G}}$, which is invertible by the row-sum bound. Its nonnegative fixed-point map sends $\mathbf{1}$ to $\boldsymbol{\gamma} \leq \mathbf{1}$, so the unique solution lies in $[0,1]^n$. Thus the tax induces efficient information use for every given precision vector.

Global optimality of information acquisition. We next verify the acquisition incentive, allowing the agent to change both her precision and her subsequent action rule. Consider an unobservable deviation to $\tau_i' \geq 0$. The other agents retain their strategies and precisions, so the following quantity is fixed during the deviation:

$$q_i = \frac{1 - \alpha_i + \sum_{j \neq i} (\widehat{g}_{ij} + \widehat{g}_{ji})\beta_j}{1 + d_i^{\text{in}}} > 0.$$

Using the posterior mean in (A.30), we obtain the deviating agent's unique optimal continuation rule:

$$a_i = \beta_i(\tau_i')x_i + (1 - \beta_i(\tau_i'))y, \qquad \beta_i(\tau_i') = \frac{\tau_i'}{\tau_i' + \tau_y} q_i.$$

For $\tau_i' > 0$, let β_i' be any candidate coefficient on the private signal. By (A.4) and (A.5), the terms in expected loss that depend on β_i' and τ_i' are

$$(1 + d_i^{\text{in}}) \left(\frac{1}{\tau_i'} + \frac{1}{\tau_y} \right) (\beta_i')^2 - \frac{2(1 + d_i^{\text{in}})q_i}{\tau_y} \beta_i' + c\tau_i'.$$

All omitted terms, including the fixed transfer, are independent of these choices. Minimizing over β_i' gives $\beta_i' = q_i\tau_i'/(\tau_i' + \tau_y)$. Substituting this coefficient and dropping a further term independent of τ_i' leaves the reduced loss

$$\frac{(1 + d_i^{\text{in}})q_i^2}{\tau_i' + \tau_y} + c\tau_i'.$$

The reduced loss extends continuously to $\tau_i' = 0$, where $\beta_i(0) = 0$, and therefore also covers no acquisition. The optimized expected payoff has first and second derivatives

$$\frac{(1 + d_i^{\text{in}})q_i^2}{(\tau_i' + \tau_y)^2} - c, \qquad -\frac{2(1 + d_i^{\text{in}})q_i^2}{(\tau_i' + \tau_y)^3} < 0.$$

The payoff is strictly concave and tends to minus infinity as $\tau_i' \to \infty$. Hence the unique global acquisition best response is

$$\tau_i' = \max\left\{0, \sqrt{\frac{1+d_i^{\text{in}}}{c}}\, q_i - \tau_y\right\}. \tag{A.31}$$

At an interior best response, substituting the optimal continuation coefficient into the first-order condition gives

$$c = (1+d_i^{\text{in}})\left(\frac{\beta_i}{\tau_i}\right)^2.$$

This condition coincides with the planner's acquisition condition (24). It also specializes (39), since $\partial^2 t_i/\partial a_i^2 = 2d_i^{\text{in}}$.

Existence, uniqueness, and implementation. It remains to show that the two incentive conditions select a unique allocation. Combining (A.31) with optimal information use gives

$$\beta_i = \max\left\{0, 1 - \sum_{j\neq i}\widetilde{g}_{ij} + \sum_{j\neq i}\widetilde{g}_{ij}\beta_j - \tau_y\sqrt{\frac{c}{1+d_i^{\text{in}}}}\right\}. \tag{A.32}$$

For $\boldsymbol{\beta} \in [0,1]^n$, the expression before subtracting $\tau_y\sqrt{\frac{c}{1+d_i^{\text{in}}}}$ lies in $(0,1]$. The map therefore takes $[0,1]^n$ into itself. Since the positive-part function is 1-Lipschitz, its contraction modulus in the supremum norm is at most

$$\max_i \sum_{j\neq i}\widetilde{g}_{ij} < 1.$$

The contraction therefore has a unique fixed point. Acquisition best responses recover the corresponding unique precision vector. The strictly concave action problems also specify optimal continuation behavior after every private acquisition deviation. Together these observations establish a unique linear equilibrium, including possible corner solutions.

For an interior solution, rearranging (A.32) gives

$$(\mathbf{I} - \widetilde{\mathbf{G}})(\mathbf{1} - \boldsymbol{\beta}) = \tau_y\sqrt{c}\,\mathbf{e}.$$

Thus $\beta_i = 1 - \tau_y\sqrt{c}\,b_i^*$, where $\mathbf{b}^* = (\mathbf{I} - \widetilde{\mathbf{G}})^{-1}\mathbf{e}$ as in Proposition 3. Choosing

$$\bar{c} = (\tau_y \max_{k\in N} b_k^*)^{-2}$$

ensures $0 < \beta_i < 1$ for every i. The corresponding precisions of private information are

$$\tau_i = \sqrt{\frac{1+d_i^{\text{in}}}{c}}\,\beta_i = \sqrt{1+d_i^{\text{in}}}\left(\frac{1}{\sqrt{c}} - \tau_y b_i^*\right).$$

These coefficients and precisions coincide with the social optimum under efficient use of information in Proposition 3. Thus the tax implements that allocation as the unique linear equilibrium. Fixed transfers remain independent of acquisition and actions and preserve the incentives at both stages.

The general acquisition condition. The following calculation establishes (39) under the regularity conditions cited in the main text. Fix the other agents' strategies and consider an interior optimal precision $\tau_i > 0$ with optimal continuation rule $a_i = \beta_i x_i + (1-\beta_i)y$. Hold this rule fixed when varying the candidate precision τ_i'. Retaining the rule is a feasible choice after any nearby acquisition deviation, so the expected payoff along these deviations, net of acquisition costs, must have a local maximum at $\tau_i' = \tau_i$. Thus its derivative, once established below, must vanish. This argument does not require differentiability of the optimized continuation value. Moreover, $\beta_i \neq 0$: otherwise reducing precision would leave actions and tax payments unchanged while lowering acquisition costs. Write $x_i = \theta + \varepsilon_i$. Conditional on the fundamental, the public signal error, and the other agents' private signal errors, define

$$f_i(z) = u_i(\beta_i(\theta+z) + (1-\beta_i)y, \mathbf{a}_{-i}, \theta; \boldsymbol{\alpha}, \mathbf{G}) - \mathcal{T}_i(\beta_i(\theta+z) + (1-\beta_i)y, \mathbf{a}_{-i}).$$

The density of ε_i at precision τ_i' is

$$\varphi_{\tau_i'}(z) = \sqrt{\frac{\tau_i'}{2\pi}}\exp(-\tau_i' z^2/2), \qquad \frac{\partial \varphi_{\tau_i'}(z)}{\partial \tau_i'} = -\frac{1}{2(\tau_i')^2}\frac{\partial^2 \varphi_{\tau_i'}(z)}{\partial z^2}.$$

For the general tax schedule, the calculation requires finite, well-defined expected payoffs. The following local conditions suffice to justify the integration steps. Choose a compact interval $I = [\underline{\tau}, \overline{\tau}] \subset (0, \infty)$ with τ_i in its interior. With primes on f_i denoting derivatives with respect to z, require

$$\mathbb{E}\left[\int_{\mathbb{R}} \sup_{\tau \in I}\left\{(|f_i(z)| + |f_i''(z)|)\varphi_\tau(z) + |f_i(z)|\left|\frac{\partial \varphi_\tau(z)}{\partial \tau}\right| + |f_i'(z)|\left|\frac{\partial \varphi_\tau(z)}{\partial z}\right|\right\} dz\right] < \infty,$$

where the expectation is over the remaining signal errors, conditional on the fundamental. Also require, for every $\tau \in I$ and almost every realization of those errors,

$$\lim_{z\to\pm\infty} f_i(z)\frac{\partial \varphi_\tau(z)}{\partial z} = 0, \qquad \lim_{z\to\pm\infty} f_i'(z)\varphi_\tau(z) = 0.$$

The integrability condition permits differentiation under the integral and interchange with the expectation over the remaining errors. The two limits eliminate the boundary terms in successive integrations by parts. Consequently,

$$\frac{\partial}{\partial \tau_i'}\int_{\mathbb{R}} f_i(z)\varphi_{\tau_i'}(z)\,dz = -\frac{1}{2(\tau_i')^2}\int_{\mathbb{R}} f_i''(z)\varphi_{\tau_i'}(z)\,dz.$$

For the quadratic tax considered here, this identity is the scalar version of Price's theorem expressed in terms of precision (Voigtlaender, 2021, Theorem 1). Because $\mathcal{T}_i(\mathbf{a}) = t_i(\mathbf{a}) + \ell_i$, the fixed transfer drops out of derivatives with respect to the action. Using $\partial^2 u_i/\partial a_i^2 = -2$ and the linear action rule gives

$$f_i''(z) = -\beta_i^2\left[2 + \frac{\partial^2 t_i(\mathbf{a})}{\partial a_i^2}\right].$$

Taking expectations over the remaining signal errors and subtracting the marginal acquisition cost gives

$$\frac{\beta_i^2}{(\tau_i')^2}\left[1 + \frac{1}{2}\mathbb{E}\left(\frac{\partial^2 t_i(\mathbf{a})}{\partial a_i^2}\right)\right] - c. \tag{A.33}$$

Evaluating this derivative at $\tau_i' = \tau_i$ and setting it equal to zero proves (39). This calculation is a necessary condition at linear optimal information use; twice continuous differentiability alone does not imply the integration conditions, linear optimal actions, or a globally optimal acquisition choice.

For the quadratic tax in (40), the normalized linear rules make f_i quadratic in the signal errors, with derivatives of lower degree. Its coefficients have finite expected absolute values because the remaining errors have finite second moments. Uniformly over the compact positive precision interval I, every term in the integrability condition is therefore bounded by an integrable random coefficient times $(1 + |z|^4)\exp(-\underline{\tau}z^2/2)$. This bound gives joint integrability, and Gaussian decay eliminates both boundary terms. Zero-precision private signals do not enter their owners' actions or these bounds. The losses are independent of the fundamental's level, so their expectations are well defined under the maintained prior. □

A.13 Proof of Proposition 6

The subsidy changes the price of private information while leaving subsequent action incentives unchanged. We first characterize equilibrium for given subsidy rates. We then show that the proposed rates implement the target allocation and satisfy the required bounds.

Acquisition incentives and implementation. Fix a subsidy vector $\mathbf{s} = (s_i)_{i \in N}$ with $s_i < 1$ for every i. If agent i makes an unobservable acquisition deviation to τ_i', the other agents retain their continuation weights. Let $q_i = 1 - \alpha_i + \sum_{j \neq i} \widehat{g}_{ij} \beta_j > 0$. As in the proof of Proposition 1, her unique optimal continuation coefficient is $\beta_i(\tau_i') = q_i \tau_i' / (\tau_i' + \tau_y)$.

To verify global optimality of acquisition, let β_i' denote a candidate coefficient on the private signal. For $\tau_i' > 0$, the terms in expected loss that depend on β_i' and τ_i' are

$$\left(\frac{1}{\tau_i'} + \frac{1}{\tau_y} \right) (\beta_i')^2 - \frac{2 q_i}{\tau_y} \beta_i' + c(1 - s_i) \tau_i'.$$

Minimizing over β_i' gives reduced loss $q_i^2 / (\tau_i' + \tau_y) + c(1 - s_i) \tau_i'$ up to an additive constant. The fixed transfer ℓ_i enters only that constant. The reduced loss extends continuously to $\tau_i' = 0$ and tends to infinity as $\tau_i' \to \infty$, because $c(1 - s_i) > 0$. The derivative of optimized expected payoff is

$$\frac{q_i^2}{(\tau_i' + \tau_y)^2} - c(1 - s_i),$$

and its second derivative is $-2 q_i^2 / (\tau_i' + \tau_y)^3 < 0$. Strict concavity therefore gives the unique global acquisition best response,

$$\tau_i' = \max \left\{ 0, \frac{q_i}{\sqrt{c(1 - s_i)}} - \tau_y \right\}.$$

At an interior choice, substitution of the optimal continuation coefficient into this first-order condition gives (43).

Combining acquisition and information use gives the componentwise fixed point

$$\beta_i = \max \left\{ 0, \, 1 - \alpha_i + \sum_{j \neq i} \widehat{g}_{ij} \beta_j - \tau_y \sqrt{c(1 - s_i)} \right\}.$$

The map takes $[0, 1]^n$ into itself and has supremum-norm contraction modulus at most $\alpha_{\max} < 1$. Its unique fixed point, together with the acquisition best responses, therefore determines a

unique linear equilibrium. If acquisition is interior, the system reduces to

$$\boldsymbol{\beta} = \mathbf{1} - \tau_y(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\big(\sqrt{c(1-s_i)}\big)_{i\in N}, \qquad \tau_i = \frac{\beta_i}{\sqrt{c(1-s_i)}}. \tag{A.34}$$

We now apply this characterization to the allocation $(\boldsymbol{\tau}^{**}, \boldsymbol{\beta}^{**})$. Lemma 3 makes this social optimum under equilibrium use of information unique and interior for sufficiently low c. The rates in (44) equate each agent's net marginal acquisition cost to her private marginal benefit at the target allocation. In particular,

$$\sqrt{c(1-s_i^{**})} = \frac{\beta_i^{**}}{\tau_i^{**}}, \qquad \frac{q_i^{**}}{\sqrt{c(1-s_i^{**})}} - \tau_y = \tau_i^{**},$$

where $q_i^{**} = 1 - \alpha_i + \sum_{j\neq i} \widehat{g}_{ij}\beta_j^{**}$. The second equality is the equilibrium information-use condition at the target allocation. Thus every target precision is the agent's global best response, and the target coefficients satisfy equilibrium information use. Once the rates are verified to lie in $[0,1)$, the contraction argument establishes implementation as the unique linear equilibrium.

Expansion and feasibility of the subsidy rates. The approximation identifies the leading subsidy and verifies feasibility for agents with positive weighted in-degree. Hold $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$ fixed. Lemma 3 gives $r_i = e_i + O(\sqrt{c})$ and $\beta_i^{**} = 1 + O(\sqrt{c})$. Its definition of r_i implies

$$\sqrt{c}\,\tau_i^{**} = \frac{1}{r_i} - \tau_y\sqrt{c} = \frac{1}{e_i} + O(\sqrt{c}).$$

Since $e_i > 0$, it follows that

$$\frac{\beta_i^{**}}{\sqrt{c}\,\tau_i^{**}} = e_i + O(\sqrt{c}).$$

Substituting into (44) gives

$$s_i^{**} = 1 - \big(e_i + O(\sqrt{c})\big)^2 = \frac{d_i^{\text{in}}}{1 + d_i^{\text{in}}} + O(\sqrt{c}),$$

which proves the expansion in (44).

For $d_i^{\text{in}} > 0$, the leading rate is strictly between zero and one. The remainder therefore leaves $s_i^{**} \in (0,1)$ for all sufficiently low c.

Agents with $d_i^{\text{in}} = 0$ require a separate argument because their leading subsidy is zero. Nonnegativity of the interaction weights implies that column i of $\widehat{\mathbf{G}}$ is zero. The information-use equations for all $k \neq i$ consequently form a closed subsystem containing neither τ_i nor β_i. Its unique solution is independent of τ_i, so $\partial\beta_k/\partial\tau_i = 0$ for $k \neq i$. A change in agent i's acquisition

changes only her own use of information. Her action enters no other agent's payoff, so the social derivative with respect to her coefficient equals her private derivative and vanishes at equilibrium information use. The term for induced changes in information use in (26) is therefore zero. The condition reduces to $c = (\beta_i^{**}/\tau_i^{**})^2$, giving $s_i^{**} = 0$ exactly.

Since the population is finite, one common low-cost bound ensures feasibility for every agent. The rates thus lie in $[0, 1)$ and are strictly positive exactly at positions with positive weighted in-degree. Together with the preceding best-response and uniqueness argument, this proves the proposition. □

A.14 Proof of Corollary 6

We first solve for the allocation induced by the simple rule. Its exact welfare expression then gives both the gain relative to equilibrium and the social value of public information.

The induced allocation. Under the simple subsidy rule, $\sqrt{c(1-s_i^0)} = \sqrt{c}\, e_i$. The interior system in (A.34) becomes

$$(\mathbf{I} - \widehat{\mathbf{G}})(\mathbf{1} - \boldsymbol{\beta}) = \tau_y \sqrt{c}\, \mathbf{e}.$$

Multiplying by $(\mathbf{I} - \widehat{\mathbf{G}})^{-1}$ gives $\boldsymbol{\beta} = \mathbf{1} - \tau_y \sqrt{c}\, \mathbf{b}^{**}$. Substituting this expression and $1/e_i = \sqrt{1 + d_i^{\text{in}}}$ into $\tau_i = \beta_i/(\sqrt{c}\, e_i)$ gives (46) and (47). These allocations are interior whenever $c < (\tau_y \max_{i \in N} b_i^{**})^{-2}$. The contraction argument in the proof of Proposition 6 establishes uniqueness.

The welfare gain. Evaluate the total loss in (A.8) at this allocation. For each agent, the contribution from private signal errors equals the acquisition cost, since $\tau_i = \beta_i/(\sqrt{c}\, e_i)$. Summing these two components gives

$$2\sqrt{c} \sum_{i \in N} \sqrt{1 + d_i^{\text{in}}}(1 - \tau_y \sqrt{c}\, b_i^{**}).$$

The loss due to errors in the public signal is $c\tau_y \mathcal{L}(\mathbf{b}^{**})$. Hence

$$\begin{aligned} W\Big(\boldsymbol{\tau}(\mathbf{s}^0), \tau_y\Big) = & -2\sqrt{c} \sum_{i \in N} \sqrt{1 + d_i^{\text{in}}} \\ & + c\tau_y \sum_{i \in N} \left[(1 - d_i^{\text{in}})(b_i^{**})^2 + \frac{2 d_i^{\text{in}}}{\sqrt{1 + d_i^{\text{in}}}}\, b_i^{**} \right], \end{aligned} \tag{A.35}$$

where the last equality uses the identity for $\mathcal{L}(\mathbf{b}^{**})$ established in the proof of Proposition 4.

Equation (A.13), together with $\sum_i(\widehat{b}_i - 1) = \sum_i d_i^{\text{in}}\widehat{b}_i$ and $\sum_i \alpha_i = \sum_i d_i^{\text{in}}$, gives

$$W(\widehat{\boldsymbol{\tau}}, \tau_y) = -\sqrt{c}\sum_{i\in N}(2 + d_i^{\text{in}}) + c\tau_y \sum_{i\in N}\left[(1 - d_i^{\text{in}})\widehat{b}_i^2 + d_i^{\text{in}}\widehat{b}_i\right].$$

Subtracting gives the expansion in (48), since

$$2 + d_i^{\text{in}} - 2\sqrt{1 + d_i^{\text{in}}} = \left(\sqrt{1 + d_i^{\text{in}}} - 1\right)^2.$$

Every summand is nonnegative. Nontrivial interaction implies that at least one agent has positive weighted in-degree, so the coefficient of $\sqrt{c}$ is strictly positive. The remaining difference is of order c for fixed $(\boldsymbol{\alpha}, \mathbf{G})$ and $\tau_y > 0$. Hence the simple rule yields a strict welfare gain for all sufficiently low c.

The social value of public information. Hold the network and acquisition cost fixed and vary τ_y within the interior region. Both $\mathbf{s}^0$ and $\mathbf{b}^{**}$ remain fixed, because they depend only on $(\boldsymbol{\alpha}, \mathbf{G})$. The induced precisions $\boldsymbol{\tau}(\mathbf{s}^0)$ and coefficients $\boldsymbol{\beta}(\mathbf{s}^0)$ nevertheless depend on τ_y, although this dependence is suppressed in the notation. Differentiation follows this induced allocation. Differentiating the exact welfare expression (A.35) gives c times the statistic in (34). As $c > 0$, the social value of public information has precisely the sign of that statistic and equals zero when the statistic vanishes. This proves the corollary. □

A.15 Fixed Transfers and Gains to Every Agent

Fixed transfers can distribute each policy's welfare gain while preserving the incentives proved above. Fix the network, the precision of public information, and an acquisition cost in the common region of the policy results. We suppress the fixed network parameters in the arguments of U_i and W. We compare each agent's expected payoff with $U_i(\widehat{\boldsymbol{\tau}}, \tau_y)$, evaluated at the equilibrium acquisition and action rules without intervention. For each instrument, we construct transfers that give every agent an equal share of the total gain and balance the planner's budget in expectation at the induced equilibrium.

In the comparisons below, U_i denotes expected payoff net of acquisition costs and before tax payments, subsidy receipts, and fixed transfers. The action rules used to evaluate it are specified for each allocation, and W is the sum of these payoffs. The fixed component ℓ_i of the agent's payment is positive when she pays the planner and negative when she receives a payment. It is chosen separately for each policy before Stage I and remains fixed after every acquisition or action deviation.

The action tax. Evaluate the payoffs at the social optimum under efficient use of information. The tax in Proposition 5 implements the action profile $\mathbf{a}^* = \sigma^*(\mathbf{x}, y; \boldsymbol{\tau}^*)$, so

$$U_i(\boldsymbol{\tau}^*, \tau_y) = \mathbb{E}[u_i(\mathbf{a}^*, \theta; \boldsymbol{\alpha}, \mathbf{G})] - c\tau_i^*.$$

Define the aggregate welfare gain from this allocation by

$$\Delta W^* = W(\boldsymbol{\tau}^*, \tau_y) - W(\widehat{\boldsymbol{\tau}}, \tau_y).$$

This gain is strictly positive. Corollary 6 supplies an allocation with a strict gain over equilibrium that is feasible for the planner who chooses both acquisition and use. The action tax implements that planner's optimum and therefore attains at least this welfare.

Expected action tax payments follow from the implemented allocation and the independent signal errors:

$$\mathbb{E}[t_i(\mathbf{a}^*)] = \sum_{j \neq i} \widehat{g}_{ji} \left[\frac{(\beta_i^*)^2}{\tau_i^*} + \frac{(\beta_j^*)^2}{\tau_j^*} + \frac{(\beta_i^* - \beta_j^*)^2}{\tau_y} \right]. \tag{A.36}$$

Deduct agent i's expected action tax from her gain before transfers, and leave her the equal share $\Delta W^*/n$. This requires the fixed payment

$$\begin{aligned} \ell_i = U_i(\boldsymbol{\tau}^*, \tau_y) - U_i(\widehat{\boldsymbol{\tau}}, \tau_y) \\ - \mathbb{E}[t_i(\mathbf{a}^*)] - \frac{\Delta W^*}{n}. \end{aligned} \tag{A.37}$$

Her expected payoff after the action tax and fixed transfer is then

$$U_i(\boldsymbol{\tau}^*, \tau_y) - \mathbb{E}[t_i(\mathbf{a}^*)] - \ell_i = U_i(\widehat{\boldsymbol{\tau}}, \tau_y) + \frac{\Delta W^*}{n} > U_i(\widehat{\boldsymbol{\tau}}, \tau_y).$$

Since W is the sum of individual payoffs, summing (A.37) gives

$$\sum_i \ell_i = \Delta W^* - \sum_i \mathbb{E}[t_i(\mathbf{a}^*)] - \Delta W^* = -\sum_i \mathbb{E}[t_i(\mathbf{a}^*)].$$

Thus $\sum_i \mathbb{E}[\mathcal{T}_i(\mathbf{a}^*)] = \sum_i \mathbb{E}[t_i(\mathbf{a}^*)] + \sum_i \ell_i = 0$. Expected tax revenue finances the fixed transfers at the induced equilibrium. Formula (A.37) determines these amounts before Stage I; the amounts remain fixed after deviations.

The acquisition subsidy. Fix either $\mathbf{s} = \mathbf{s}^{**}$ or $\mathbf{s} = \mathbf{s}^{0}$. Evaluate payoffs at the induced precision vector $\boldsymbol{\tau}(\mathbf{s})$ and the corresponding equilibrium action rules $\sigma^{\text{eq}}(\mathbf{x}, y; \boldsymbol{\tau}(\mathbf{s}))$. Thus

$$U_i(\boldsymbol{\tau}(\mathbf{s}), \tau_y) = \mathbb{E}[u_i(\sigma^{\text{eq}}(\mathbf{x}, y; \boldsymbol{\tau}(\mathbf{s})), \theta; \boldsymbol{\alpha}, \mathbf{G})] - c\tau_i(\mathbf{s}).$$

Define the aggregate welfare gain under the subsidy vector $\mathbf{s}$ by

$$\Delta W(\mathbf{s}) = W(\boldsymbol{\tau}(\mathbf{s}), \tau_y) - W(\widehat{\boldsymbol{\tau}}, \tau_y).$$

For the simple subsidy rule, Corollary 6 gives $\Delta W(\mathbf{s}^0) > 0$. The implementing subsidy attains the social optimum under equilibrium use of information. Since the simple rule's allocation is feasible for that planner, $\Delta W(\mathbf{s}^{**}) > 0$ as well. To leave each agent the equal share $\Delta W(\mathbf{s})/n$, including her subsidy receipt in the transfer calculation, choose

$$\begin{aligned} \ell_i = U_i(\boldsymbol{\tau}(\mathbf{s}), \tau_y) - U_i(\widehat{\boldsymbol{\tau}}, \tau_y) \\ + cs_i\tau_i(\mathbf{s}) - \frac{\Delta W(\mathbf{s})}{n}. \end{aligned} \tag{A.38}$$

The expected payoff after the subsidy and fixed transfer is

$$U_i(\boldsymbol{\tau}(\mathbf{s}), \tau_y) + cs_i\tau_i(\mathbf{s}) - \ell_i = U_i(\widehat{\boldsymbol{\tau}}, \tau_y) + \frac{\Delta W(\mathbf{s})}{n} > U_i(\widehat{\boldsymbol{\tau}}, \tau_y).$$

Summing (A.38) and using the definition of the aggregate gain gives

$$\sum_i \ell_i = \Delta W(\mathbf{s}) + c\sum_i s_i\tau_i(\mathbf{s}) - \Delta W(\mathbf{s}) = c\sum_i s_i\tau_i(\mathbf{s}).$$

The fixed payments therefore finance all subsidies at the induced equilibrium:

$$\sum_i \mathcal{T}_i(\tau_i(\mathbf{s})) = \sum_i \ell_i - c\sum_i s_i\tau_i(\mathbf{s}) = 0.$$

After a deviation, the fixed payments stay unchanged and the subsidy continues to reimburse the agent's actual acquisition choice.

Under both instruments, unrestricted enforceable fixed transfers and the absence of liquidity constraints permit this redistribution without changing marginal incentives. Every agent receives the same strictly positive ex ante gain relative to equilibrium without intervention. The action tax balances the budget in expectation, since realized revenue varies with the signals. The constructions require neither budget balance after deviations nor a gain to each agent in every signal realization. □

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